BMI and documented hypercapnia in acute care

Expanded scientific report for investigator review

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Protected research draft. Investigator interpretation and scientific approval remain pending.

Saved run: curve-analyze-20261024T022943-53435. Analysis: bmi_hypercapnia_rfs_curve_audit_v6.

Source: identity_bound_source_declarations_passed; analysis completed: TRUE. Review states in the immutable bundle are recorded at estimation time; later signed review receipts are separate.

1 Introduction

We describe BMI–hypercapnia relationships within a selected acute-care cohort while exposing how historical source-entry pathways shape the observed population. These pathways record representation in reason-for-suspicion (RFS) source files; they do not necessarily identify every clinical criterion present at an encounter. Separating entry routes can change the observed association itself and does not eliminate outcome-dependent selection.

2 Methods

2.1 Source, eligibility and fixed index

The authenticated original FULL_DATA encounter list defines candidate visits. Adults had a 2022 emergency or inpatient encounter and available valid BMI ≥30 kg/m². Age uses the inherited birth-year proxy. Each patient contributes the first qualifying encounter before gas availability and adjustment completeness. Earlier unre-

solved eligibility, conflicting evidence and earliest-date ties remain unresolved. Later encounters never replace an obesity-only index.

BMI follows the approved encounter/prior reported-value hierarchy, followed by calculation from observed height and weight, with inherited rounding and validity rules. One-unit diagnostic bins around BMI 40 and 50 describe measurement and selection; they are not a regression-discontinuity design. No statistical BMI imputation is performed. Available BMI is not necessarily contemporaneous with the outcome. The audit reports measurement methods, lags, unknown chronology and consequences of the existing rules. Signed lags equal encounter-start date minus measurement date: positive means prior, zero means the same calendar day, and negative means later during the encounter. Component gaps are absolute differences between height and weight dates.

2.2 Historical source-entry populations

A confirmed positive ABG, VBG, RESPFAIL, VENTSUPPORT or PREDISPOSITION source flag establishes the primary population, including encounters also represented through OBESITY. Obesity-only membership requires OBESITY and authenticated zeros in all five other families. Obesity-only indexes are excluded from scientific comparisons: removing the gas-related entry pathways selects against the evidence needed to assess the physiological outcomes. Their counts remain in the flow for reconciliation; source discrepancies and descriptive profiles are confined to the audit. Unknown flags remain unknown. All-six-zero discrepancies require source investigation. RFS metadata are applied after the common index is selected.

The combined indexed cohort is a reconciliation reference. Non-OBESITY-only, overlapping and unknown-OBESITY/confirmed-non-OBESITY subgroups are descriptive. BMI ≥40 remains a descriptive stratum and is not equated with obesity-route membership. An overlapping all-OBESITY supporting analysis is confined to the audit.

2.3 Outcomes and adjustment

Arterial hypercapnia requires ordinary authenticated ABG $PaCO_2$ ≥45 mm Hg. Combined gas accepts ABG $PaCO_2$ ≥45 or VBG $PvCO_2$ ≥50 mm Hg; both specimens are not required. No pH restriction is applied to the established primary outcomes. Strict validity limits are 5<arterial<200 and 10<venous<250 mm Hg, with reviewed units. Available-evidence negatives do not establish that an unmeasured specimen is normal. Unusable or unavailable outcomes remain unknown.

The ICD composite comprises J96.02, J96.12, J96.22, J96.92 and E66.2, with captured absence negative only under complete diagnosis capture. E66.2 introduces obesity-related ascertainment. Index-calendar-day gas sensitivities use recorded-result date equal to encounter-start date, retaining the index. This export does not authenticate collection times.

Adjustment retains linear age and the inherited sex, combined race/ethnicity and geographic-region coding. Completeness means nonmissing values for inherited analysis variables, including retained Unknown categories. The White reference category includes non-Hispanic or unknown ethnicity; inherited race-category precedence is unchanged. Raw race × ethnicity cross-tabs are shown only when authenticated.

2.4 Models, probabilities and uncertainty

Primary models use type-7 BMI quantiles .05/.35/.65/.95 from the primary adjustment-complete population: two interior knots, two boundaries and three natural-spline basis columns. The basis remains fixed across primary outcomes, timing, matched analyses and predictions. The primary current-measurement sensitivity reuses this basis. The overlapping all-OBESITY supporting analysis uses its own basis and is reported in the audit. Obesity-only and obesity-only current-measurement models are not registered. Constant outcomes, duplicate knots, separation, rank and convergence failures in planned models remain explicit without estimator substitution.

Gas curves standardize over each model's outcome-evaluable population; ICD uses known capture within group. Observation curves describe usable gas-evidence availability. Complete diagnosis capture is reported as a source fact, not as ICD positivity or a constant availability curve. No physiological outcomes are projected to untested

patients. Fixed BMI 45 versus 40 contrasts require training-range support. Local counts and marginal covariate coverage expose sparsity even inside the global BMI range; these checks do not establish joint covariate support.

Pointwise 95% intervals use full coefficient covariance and condition on the empirical reference and fixed spline basis. They omit uncertainty from selection, measurement error and unmeasured confounding. Apparent calibration is in-sample; fitted/observed mean agreement is not independent validation.

2.5 Secondary scenario and supporting checks

The named secondary scenario subtracts exactly one BMI unit when observed BMI >30; BMI=30 is unchanged. There is no floor or reselection. Paired predictions hold index, RFS membership and other covariates fixed. Reported quantities are observed and shifted mean probabilities, their signed difference and PIF=1−p_shifted/p_observed. Signed original-BMI-band contributions and their full covariance reproduce the total difference and variance. Marginal curve-density integration is not used.

Same-sample linear-BMI checks and matched ICD/gas analyses remain registered. A limited current-measurement sensitivity uses the already-selected BMI when reported at the index or calculated from height and weight both authenticated to that encounter. Full-cohort MAR/MNAR projections and IPW are excluded. The report is rendered from saved aggregates without clinical reads or refitting.

3 Results

3.1 Primary cohort and usable evidence

Table 1: Fixed-index RFS partition

population id	n	complete	incomplete	denominator	status
primary	176922	176922	0	223514	assessed
obesity only	46592	46592	0	223514	assessed
unresolved	0	0	0	223514	assessed

Source-entry partition at the fixed index

Mutually exclusive patient groups; overlap remains in primary

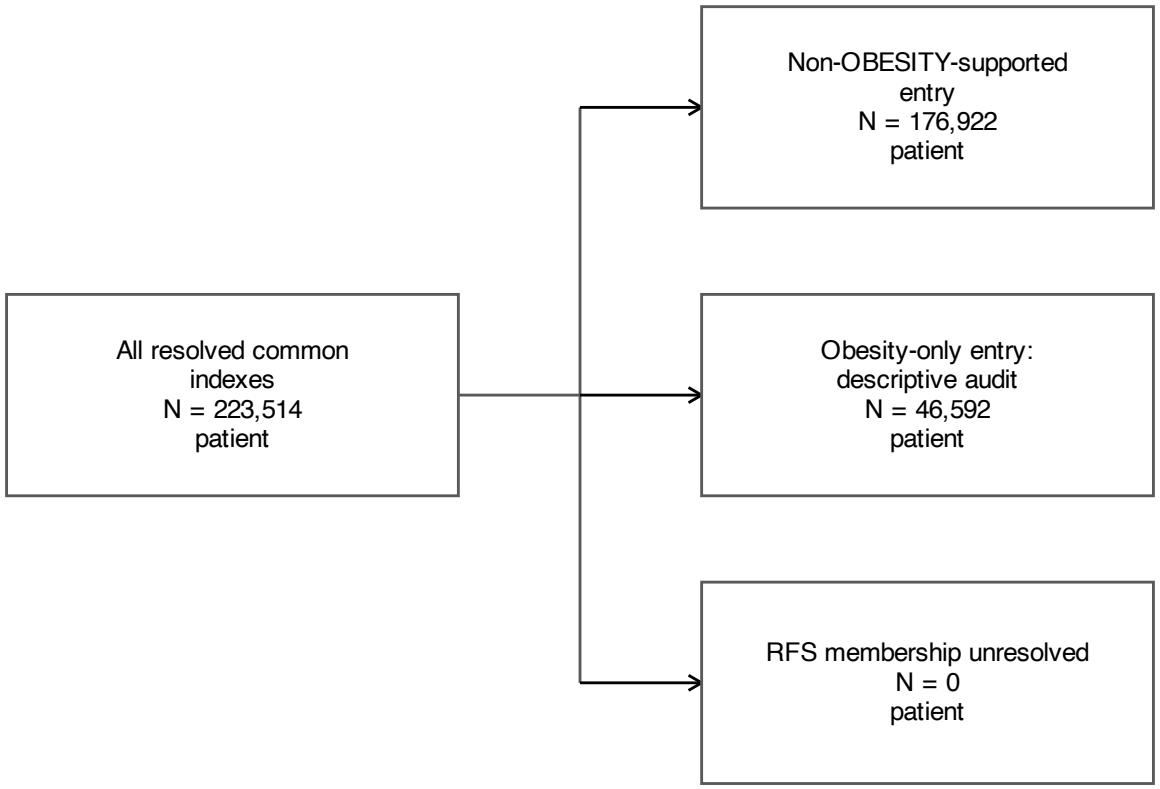


Figure 1: Fixed common indexes partition into primary non-OBESITY-supported source entry, obesity-only source entry and unresolved RFS. Overlapping OBESITY/non-OBESITY membership remains primary. Counts and units derive from the ledger; all-OBESITY supporting models overlap and are not a fourth partition branch.

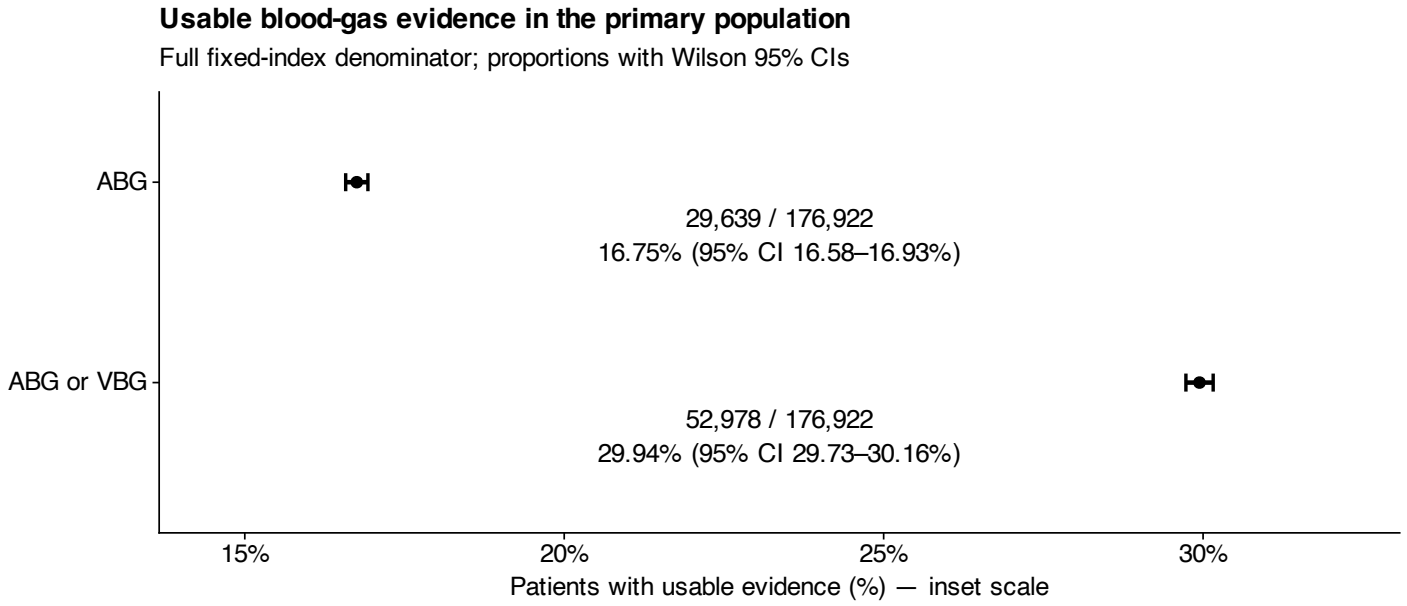


Figure 2: Encounter-wide usable gas evidence among all primary fixed indexes, before adjustment completeness. Labels give evaluable/full N, percentage and 95% Wilson CI; capped intervals use their exact endpoints on a labeled inset scale. ABG: arterial blood gas; VBG: venous blood gas. The combined row accepts either usable specimen; the two rows overlap and are not summed. These percentages describe evidence availability, not hypercapnia positivity. Primary includes non-OBESITY pathways and their OBESITY overlaps. Complete diagnosis capture, when present, is a source fact rather than ICD positivity and is reported separately. Obesity-only counts and residual-gas discrepancies are confined to the cohort flow and audit; they do not define a scientific comparator or weighting analysis.

Diagnosis capture was known for 176,922 of 176,922 primary indexes. This describes source coverage, not the presence of an ICD hypercapnia code; observed ICD positivity is reported below. The complete source-entry reconciliation and residual-gas audit are in the companion audit report.

3.2 Source-entry population: primary

grain: patient; status: assessed; analysis population id: primary.

Table 2: Population and evidence accounting

population id	label	n records	n encounters	n patients	n
Q	Resolved first indexes (Q)	176922	176922	176922	176922
missing adjustment	missing adjustment	0	0	0	0
C	Complete adjustment (C)	176922	176922	176922	176922
C icd	Known diagnosis capture (C icd)	176922	176922	176922	176922
C abg encounter	C abg encounter	29639	29639	29639	29639
C any gas encounter	C any gas encounter	52978	52978	52978	52978

variable_id=age_at_encounter

n missing: 0; status: assessed; minimum: 18; q95: 82; q99: 87; n invalid: 0; analysis population id: primary.

Table 3: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	q01	q05
Q	176922	176922	77	57.47882	16.77562	21	27
C	176922	176922	77	57.47882	16.77562	21	27
C abg encounter	29639	29639	75	61.57765	14.93598	23	32
C any gas encounter	52978	52978	76	59.75945	15.68863	22	30

Table 4: Participant profiles — panel 2/2

population id	q25	median	q75	maximum
Q	45	60	71	96
C	45	60	71	96
C abg encounter	53	64	73	92
C any gas encounter	50	62	72	96

variable_id=bmi

n missing: 0; status: assessed; minimum: 30; q01: 30; q95: 49.7000007629395; maximum: 89.8000030517578; n invalid: 0; analysis population id: primary.

Table 5: Participant profiles — panel 1/2

population id	n	n observed	n distinct	mean	sd	q05	q25
Q	176922	176922	488	37.903	6.699107	30.4	32.6
C	176922	176922	488	37.903	6.699107	30.4	32.6
C abg encounter	29639	29639	436	37.01368	6.687356	30.3	32.1
C any gas encounter	52978	52978	458	37.12698	6.652777	30.3	32.2

Table 6: Participant profiles — panel 2/2

population id	median	q75	q99
Q	36.3	41.6	60.5
C	36.3	41.6	60.5
C abg encounter	35	39.9	61.3
C any gas encounter	35.2	40.1	60.8

variable_id=sex

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 7: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	176922	176922	0	2	assessed	primary
C	176922	176922	0	2	assessed	primary
C abg encounter	29639	29639	0	2	assessed	primary
C any gas encounter	52978	52978	0	2	assessed	primary

variable_id=race_ethnicity

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 8: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	176922	176922	0	7	assessed	primary
C	176922	176922	0	7	assessed	primary
C abg encounter	29639	29639	0	7	assessed	primary
C any gas encounter	52978	52978	0	7	assessed	primary

variable_id=location

Unavailable throughout this group: mean, sd, minimum, q01, q05, q25, median, q75, q95, q99, maximum, n_invalid.

Table 9: Participant profiles

population id	n	n observed	n missing	n distinct	status	analysis population id
Q	176922	176922	0	4	assessed	primary
C	176922	176922	0	4	assessed	primary
C abg encounter	29639	29639	0	4	assessed	primary
C any gas encounter	52978	52978	0	4	assessed	primary

Arterial PCO_2 ≥45 mm Hg: 15,541 documented-positive outcomes among 29,639 model patients; observed positivity 52.43% (95% CI 51.87–53%) (Wilson interval). Model status: successful.

Arterial ≥45 or venous ≥50 mm Hg: 23,621 documented-positive outcomes among 52,978 model patients; observed positivity 44.59% (95% CI 44.16–45.01%) (Wilson interval). Model status: successful.

Documented ICD composite: 8,634 documented-positive outcomes among 176,922 model patients; observed positivity 4.88% (95% CI 4.781–4.981%) (Wilson interval). Model status: successful.

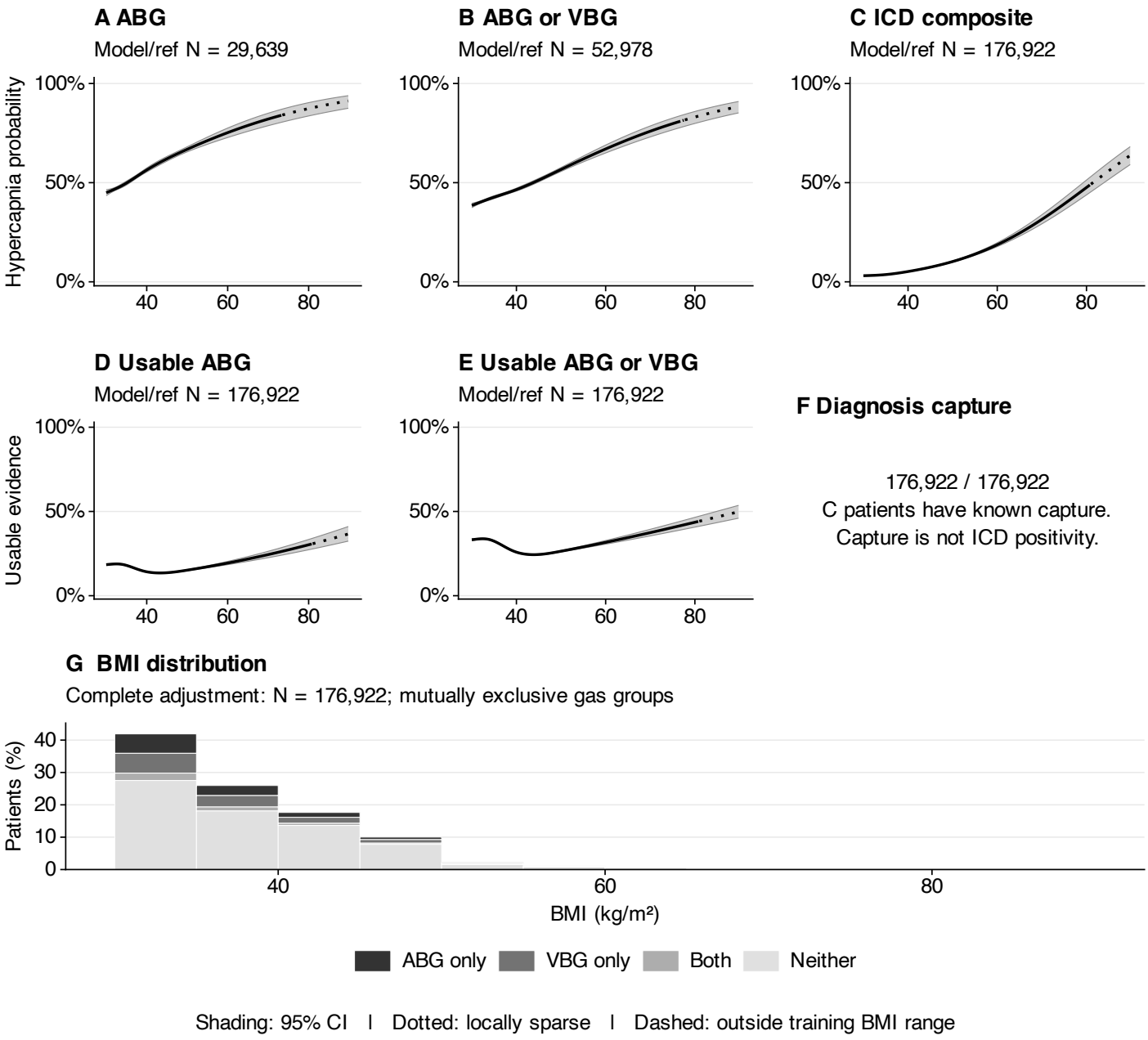


Figure 3: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred.

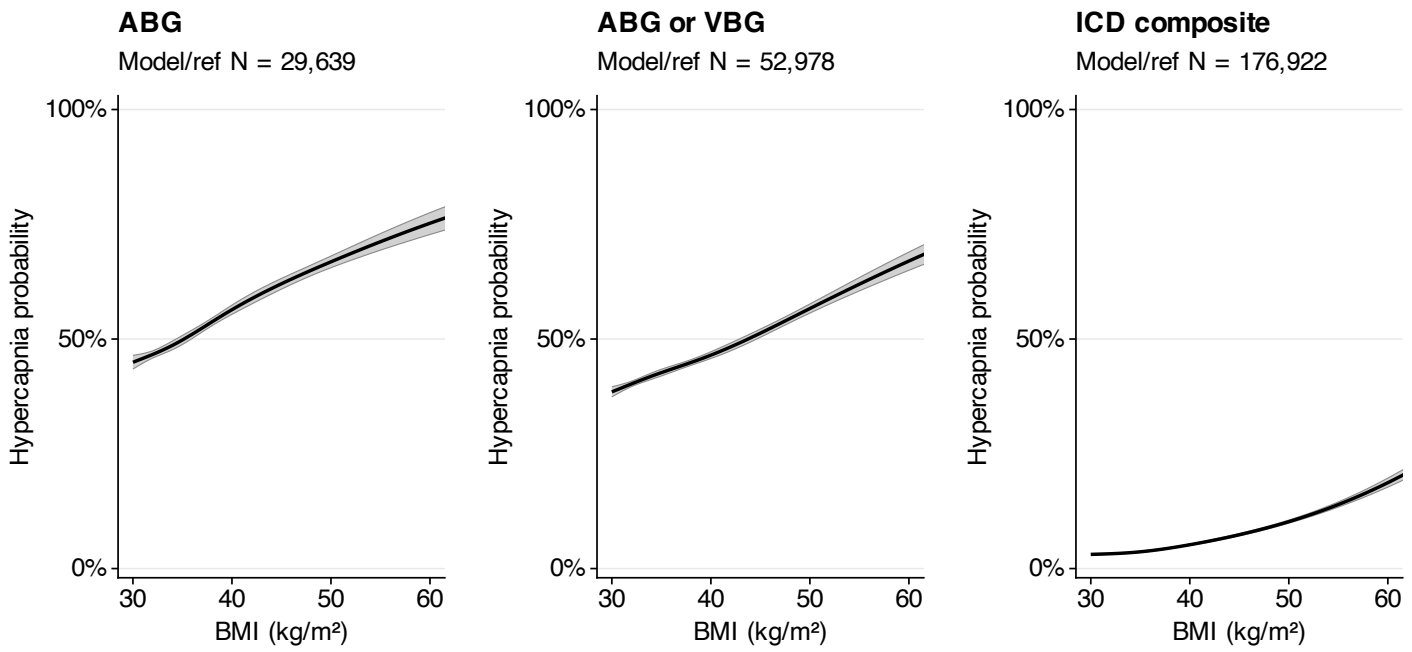


Figure 4: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F1. BMI–hypercapnia relationships at the first eligible adult 2022 acute-care encounter. ABG denotes $PaCO_2$ ≥45 mm Hg; ABG or VBG denotes $PaCO_2$ ≥45 or $PvCO_2$ ≥50 mm Hg. ICD denotes the five-code composite, including E66.2. Gas curves standardize over outcome-evaluable patients; ICD uses known diagnosis capture. Model/ref N is the common model and reference count. Availability and BMI frequency use all adjustment-complete patients (C); the four frequency groups are mutually exclusive. Adjustment: age, sex, race/ethnicity and region with a fixed natural BMI spline. Shading and fine boundaries show pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Open availability points use Wilson intervals when no model is estimable. Known capture is distinct from ICD positivity. ABG/VBG: arterial/venous blood gas. Conditional model uncertainty excludes selection and measurement assumptions; no population transport or causal effect is inferred. This labeled detail view limits the displayed BMI axis to 30–60; no observations were removed from estimation.

bmi low: 40; bmi high: 45; status: estimated; reason: ; analysis population id: primary; population id: primary; ascertainment window: encounter; model form: spline.

Table 10: Supported BMI 45 versus 40 contrasts — panel 1/3

model id	reference id	n reference	n model	probability low bmi	probability high bmi	probability low bmi low	probability low bmi high
abg/encounter/spline	C abg encounter	29639	29639	0.563898	0.6204125	0.5538255	0.5739181
any gas/encounter/spline	C any gas encounter	52978	52978	0.4649322	0.5127818	0.4575954	0.4722842
icd/encounter/spline	C icd	176922	176922	0.05191386	0.07372897	0.05038018	0.0534916

Table 11: Supported BMI 45 versus 40 contrasts — panel 2/3

model id	probability high bmi low	probability high bmi high	difference per1000	conf low per1000	conf high per1000	prevalence ratio	ratio low
abg/encounter/spline	0.6086704	0.6320152	56.51448	50.37713	62.65183	1.100221	1.089214
any gas/encounter/spline	0.5038924	0.521663	47.84958	43.14841	52.55074	1.102917	1.092791
icd/encounter/spline	0.07118096	0.07636069	21.81511	20.24304	23.38718	1.420217	1.392341

Table 12: Supported BMI 45 versus 40 contrasts — panel 3/3

model id	ratio high	reference population id	outcome id
abg/encounter/spline	1.111339	C abg encounter	abg
any gas/encounter/spline	1.113138	C any gas encounter	any gas
icd/encounter/spline	1.448652	C icd	icd

Arterial PCO_2 ≥45 mm Hg: standardized probability at BMI 40 was 56.39% and at BMI 45 was 62.04%. The difference was 5.651 (5.038, 6.265) percentage points (95% CI), within C_abg_encounter. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

Arterial ≥45 or venous ≥50 mm Hg: standardized probability at BMI 40 was 46.49% and at BMI 45 was 51.28%. The difference was 4.785 (4.315, 5.255) percentage points (95% CI), within C_any_gas_encounter. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

Documented ICD composite: standardized probability at BMI 40 was 5.191% and at BMI 45 was 7.373%. The difference was 2.182 (2.024, 2.339) percentage points (95% CI), within C_icd. This is a fixed-BMI contrast, not the one-unit distribution-shift PIF.

quantile type: 7; status: resolved; reference id: C; basis: splines::ns; intercept FALSE; 3 columns; analysis population id: primary; reference population id: C.

Table 13: Saved group-specific numeric basis

basis id	knot probability	knot	kind
026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105	0.05	30.4	boundary
026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105	0.35	33.9	interior
026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105	0.65	39.3	interior

basis id	knot probability	knot	kind
026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105	0.95	49.7	boundary

Table 14: Encounter-wide spline fit status

Field	Value
model id	abg/encounter/spline
family	primary
outcome id	abg
window	encounter
training population id	C abg encounter
field	abg
model form	spline
source status	qualified
formula text	.outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location
basis id	026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105
weight specification	none
covariance method	model-based binomial; empirical reference fixed
n eligible	176922
n excluded outcome	147283
n excluded adjustment	0
n model	29639
n patients	29639
n events	15541
n nonevents	14098
n parameters	15
model rank	15
status	successful
failure reason	
warnings	
fit runtime	0.161
analysis population id	primary
ascertainment window	encounter
population id	primary
source reason	unavailable

Table 15: Complete primary coefficients

Term	Estimate	SE	95% CI	P value	Interpretation
(Intercept)	-0.7443513	0.06549514	-0.7444 (-0.8727, -0.616)	6.247462e-30	log-odds contrast
bmi ns1	0.4774925	0.04381378	0.4775 (0.3916, 0.5634)	1.175264e-27	basis coefficient; no standalone clinical OR
bmi ns2	0.8627809	0.06496352	0.8628 (0.7355, 0.9901)	2.983564e-40	basis coefficient; no standalone clinical OR
bmi ns3	0.8146604	0.03563716	0.8147 (0.7448, 0.8845)	1.166056e-115	basis coefficient; no standalone clinical OR
age at encounter	0.00650236	8.20106e-04	0.006502 (0.004895, 0.00811)	2.214828e-15	log-odds contrast
sex1	0.3179027	0.02403412	0.3179 (0.2708, 0.365)	6.117298e-40	log-odds contrast
race ethnicity1	-0.110524	0.03192594	-0.1105 (-0.1731, -0.04795)	5.364e-04	log-odds contrast
race ethnicity2	-0.1190858	0.04802947	-0.1191 (-0.2132, -0.02495)	0.0131592	log-odds contrast
race ethnicity3	-0.1713243	0.1225482	-0.1713 (-0.4115, 0.06887)	0.1621085	log-odds contrast

Term	Estimate	SE	95% CI	P value	Interpretation
race ethnicity4	-0.139982	0.1188708	-0.14 (-0.373, 0.093)	0.2389568	log-odds contrast
race ethnicity5	-0.1841403	0.2568364	-0.1841 (-0.6875, 0.3192)	0.4734015	log-odds contrast
race ethnicity6	-0.02230447	0.0486884	-0.0223 (-0.1177, 0.07312)	0.6468759	log-odds contrast
location1	0.1168835	0.03160167	0.1169 (0.05495, 0.1788)	2.167504e-04	log-odds contrast
location2	0.04044335	0.03783289	0.04044 (-0.03371, 0.1146)	0.2850698	log-odds contrast
location3	-0.02878643	0.03477898	-0.02879 (-0.09695, 0.03938)	0.4078425	log-odds contrast
sex=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
race ethnicity=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
location=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level

model id: abg/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: abg; ascertainment window: encounter; model form: spline.

Table 16: Apparent fit diagnostics

metric	value
log likelihood	-20102.16
null deviance	41018.1
residual deviance	40204.33
AIC	40234.33
Brier	0.2426885
log loss	0.6782335
iterations	4
residual df	29624
scaled condition number	9.720362
events per parameter	1036.067
nonevents per parameter	939.8667

Table 17: Encounter-wide spline fit status

Field	Value
model id	any gas/encounter/spline
family	primary
outcome id	any gas
window	encounter
training population id	C any gas encounter
field	any gas
model form	spline
source status	qualified
formula text	.outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location
basis id	026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105
weight specification	none
covariance method	model-based binomial; empirical reference fixed
n eligible	176922
n excluded outcome	123944
n excluded adjustment	0
n model	52978
n patients	52978
n events	23621
n nonevents	29357
n parameters	15

Field	Value
model rank	15
status	successful
failure reason	
warnings	
fit runtime	0.236
analysis population id	primary
ascertainment window	encounter
population id	primary
source reason	unavailable

Table 18: Complete primary coefficients

Term	Estimate	SE	95% CI	P value	Interpretation
(Intercept)	-1.303706	0.04848148	-1.304 (-1.399, -1.209)	2.812911e-159	log-odds contrast
bmi ns1	0.2784644	0.03225256	0.2785 (0.2153, 0.3417)	5.931061e-18	basis coefficient; no standalone clinical OR
bmi ns2	0.6974113	0.04972655	0.6974 (0.5999, 0.7949)	1.097228e-44	basis coefficient; no standalone clinical OR
bmi ns3	0.6437564	0.02590718	0.6438 (0.593, 0.6945)	2.679285e-136	basis coefficient; no standalone clinical OR
age at encounter	0.01352009	5.983956e-04	0.01352 (0.01235, 0.01469)	4.975362e-113	log-odds contrast
sex1	0.2718704	0.01806465	0.2719 (0.2365, 0.3073)	3.459564e-51	log-odds contrast
race ethnicity1	-0.04276667	0.02336648	-0.04277 (-0.08856, 0.003031)	0.0672115	log-odds contrast
race ethnicity2	-0.08115003	0.03653968	-0.08115 (-0.1528, -0.009534)	0.02635946	log-odds contrast
race ethnicity3	-0.2250825	0.0963428	-0.2251 (-0.4139, -0.03625)	0.01947734	log-odds contrast
race ethnicity4	-0.4527456	0.08747207	-0.4527 (-0.6242, -0.2813)	2.268291e-07	log-odds contrast
race ethnicity5	-0.1230116	0.2138289	-0.123 (-0.5421, 0.2961)	0.5651018	log-odds contrast
race ethnicity6	-0.1907534	0.03253794	-0.1908 (-0.2545, -0.127)	4.559761e-09	log-odds contrast
location1	-0.05575314	0.02172421	-0.05575 (-0.09833, -0.01317)	0.01027584	log-odds contrast
location2	-0.1264609	0.02926886	-0.1265 (-0.1838, -0.06909)	1.555619e-05	log-odds contrast
location3	-0.1643927	0.02879889	-0.1644 (-0.2208, -0.1079)	1.141105e-08	log-odds contrast
sex=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
race ethnicity=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
location=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level

model id: any_gas/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: any_gas; ascertainment window: encounter; model form: spline.

Table 19: Apparent fit diagnostics

metric	value
log likelihood	-35618.53

metric	value
null deviance	72820.84
residual deviance	71237.06
AIC	71267.06
Brier	0.2398443
log loss	0.6723268
iterations	4
residual df	52963
scaled condition number	9.58629
events per parameter	1574.733
nonevents per parameter	1957.133

Table 20: Encounter-wide spline fit status

Field	Value
model id	icd/encounter/spline
family	primary
outcome id	icd
window	encounter
training population id	C icd
field	icd
model form	spline
source status	qualified
formula text	.outcome ~ bmi ns1 + bmi ns2 + bmi ns3 + age at encounter + sex + race ethnicity + location
basis id	026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105
weight specification	none
covariance method	model-based binomial; empirical reference fixed
n eligible	176922
n excluded outcome	0
n excluded adjustment	0
n model	176922
n patients	176922
n events	8634
n nonevents	168288
n parameters	15
model rank	15
status	successful
failure reason	
warnings	
fit runtime	0.777
analysis population id	primary
ascertainment window	encounter
population id	primary
source reason	unavailable

Table 21: Complete primary coefficients

Term	Estimate	SE	95% CI	P value	Interpretation
(Intercept)	-5.447039	0.06739039	-5.447 (-5.579, -5.315)	0	log-odds contrast
bmi ns1	0.5966077	0.03621438	0.5966 (0.5256, 0.6676)	5.610631e-61	basis coefficient; no standalone clinical OR
bmi ns2	1.092816	0.07349178	1.093 (0.9488, 1.237)	5.168485e-50	basis coefficient; no standalone clinical OR
bmi ns3	1.214574	0.02970283	1.215 (1.156, 1.273)	0	basis coefficient; no standalone clinical OR
age at encounter	0.03166113	7.927538e-04	0.03166 (0.03011, 0.03321)	0	log-odds contrast
sex1	0.1788576	0.02292163	0.1789 (0.1339, 0.2238)	6.044934e-15	log-odds contrast
race ethnicity1	-0.1356647	0.03149877	-0.1357 (-0.1974, -0.07393)	1.654956e-05	log-odds contrast

Term	Estimate	SE	95% CI	P value	Interpretation
race ethnicity2	0.0357209	0.05098053	0.03572 (-0.0642, 0.1356)	0.4835045	log-odds contrast
race ethnicity3	0.3354555	0.1449119	0.3355 (0.05143, 0.6195)	0.02061882	log-odds contrast
race ethnicity4	-0.08033069	0.1428102	-0.08033 (-0.3602, 0.1996)	0.5737757	log-odds contrast
race ethnicity5	0.227353	0.2463188	0.2274 (-0.2554, 0.7101)	0.3560055	log-odds contrast
race ethnicity6	0.1810108	0.0402531	0.181 (0.1021, 0.2599)	6.897807e-06	log-odds contrast
location1	-0.04176757	0.03056206	-0.04177 (-0.1017, 0.01813)	0.1717358	log-odds contrast
location2	-0.1945384	0.03710537	-0.1945 (-0.2673, -0.1218)	1.581044e-07	log-odds contrast
location3	0.06972893	0.0311753	0.06973 (0.008626, 0.1308)	0.02530779	log-odds contrast
sex=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
race ethnicity=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level
location=0	unavailable	unavailable	Reference; not estimated	unavailable	reference level

model id: icd/encounter/spline; note: apparent in-sample diagnostic; no validation or causal claim; analysis population id: primary; population id: primary; outcome id: icd; ascertainment window: encounter; model form: spline.

Table 22: Apparent fit diagnostics

metric	value
log likelihood	-32763.07
null deviance	68989
residual deviance	65526.14
AIC	65556.14
Brier	0.0453141
log loss	0.1851837
iterations	6
residual df	176907
scaled condition number	8.864508
events per parameter	575.6
nonevents per parameter	11219.2

Maximum decrement: 1.

Table 23: Secondary observed-population BMI scenario — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	29639	29639	15541	29206	0.9853909	98.54
any gas/encounter/spline	C any gas encounter	52978	52978	23621	52232	0.9859187	98.59
icd/encounter/spline	C icd	176922	176922	8634	174658	0.9872034	98.72

Table 24: Secondary observed-population BMI scenario — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	52.43 (51.87, 52.99)	51.39 (50.79, 51.99)	10.42 (8.379, 12.47)	1.988 (1.597, 2.378)
any gas/encounter/spline	44.59 (44.17, 45)	43.74 (43.3, 44.18)	8.512 (7.054, 9.97)	1.909 (1.581, 2.236)

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
icd/encounter/spline	4.88 (4.782, 4.98)	4.619 (4.521, 4.718)	2.615 (2.337, 2.893)	5.359 (4.796, 5.918)

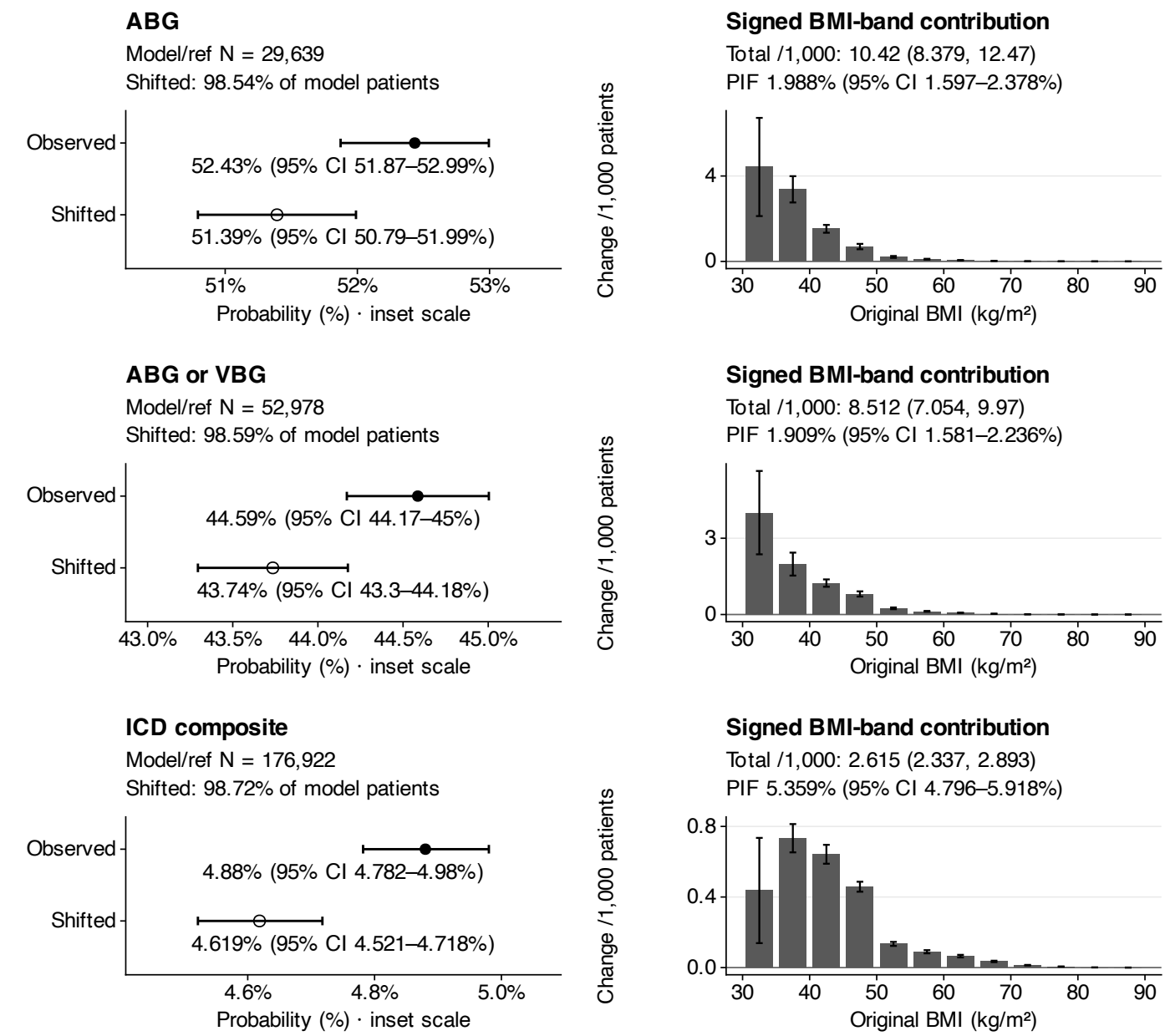


Figure 5: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F2. Paired scenario: One-unit BMI reduction for observed BMI >30; no floor. ABG: arterial PCO_2 ≥45 mm Hg; ABG or VBG: arterial ≥45 or venous ≥50 mm Hg; ICD: five-code composite including E66.2. Left: observed and shifted standardized probabilities with model-based 95% CIs and printed limits. Left probability axes use different labeled inset scales; compare printed values across outcomes. N identifies the common model/reference population; shifted percentage is a descriptive cohort fraction. Right: signed original-BMI-band contributions per 1,000 reference patients, with 95% CIs. Total-difference/PIF limits are printed; contribution covariance reproduces total variance. Right axes vary by outcome and retain zero and all limits. Gas samples are outcome-evaluable; ICD uses known capture. Adjustment, source-entry membership and covariates are fixed. PIF is the model-standardized potential impact fraction. These conditional scenarios do not estimate a weight-loss treatment effect.

fit bmi min: 30; fit bmi max: 89.8000030517578; n observed outside: 0; maximum observed gap: 0; n unseen categories: 0; analysis population id: primary; population id: primary.

Table 25: Curve-grid and transformed support diagnostics — panel 1/2

model id	n reference	n transformed outside	pct transformed outside	maximum transformed gap	reference id	policy id	method
abg/encounter/spline	29639	3181	10.73248	0.8999996	C abg encounter	bmi gt30 minus1 v1	observed
any gas/encounter/spline	52978	5548	10.47227	0.8999996	C any gas encounter	bmi gt30 minus1 v1	observed
icd/encounter/spline	176922	15514	8.768836	0.8999996	C icd	bmi gt30 minus1 v1	observed
abg/index calendar day/spline	17316	1882	10.86856	0.8999996	C abg index calendar day	bmi gt30 minus1 v1	observed
any gas/index calendar day/spline	38877	4041	10.39432	0.8999996	C any gas index calendar day	bmi gt30 minus1 v1	observed
matched/abg/icd	29639	3181	10.73248	0.8999996	M abg	bmi gt30 minus1 v1	observed
matched/abg/abg	29639	3181	10.73248	0.8999996	M abg	bmi gt30 minus1 v1	observed
matched/any gas/icd	52978	5548	10.47227	0.8999996	M any gas	bmi gt30 minus1 v1	observed
matched/any gas/any gas	52978	5548	10.47227	0.8999996	M any gas	bmi gt30 minus1 v1	observed
abg/encounter/spline	29639	0	0	0	C abg encounter	none; relationship	fixed BMI curve
abg/encounter/linear	29639	0	0	0	C abg encounter	none; relationship	fixed BMI curve
any gas/encounter/spline	52978	0	0	0	C any gas encounter	none; relationship	fixed BMI curve
any gas/encounter/linear	52978	0	0	0	C any gas encounter	none; relationship	fixed BMI curve
icd/encounter/spline	176922	0	0	0	C icd	none; relationship	fixed BMI curve
icd/encounter/linear	176922	0	0	0	C icd	none; relationship	fixed BMI curve
abg/index calendar day/spline	17316	0	0	0	C abg index calendar day	none; relationship	fixed BMI curve
abg/index calendar day/linear	17316	0	0	0	C abg index calendar day	none; relationship	fixed BMI curve
any gas/index calendar day/spline	38877	0	0	0	C any gas index calendar day	none; relationship	fixed BMI curve
any gas/index calendar day/linear	38877	0	0	0	C any gas index calendar day	none; relationship	fixed BMI curve
observation/abg/encounter	176922	0	0	0	C	none; relationship	fixed BMI curve
observation/vbg/encounter	176922	0	0	0	C	none; relationship	fixed BMI curve
observation/any gas/encounter	176922	0	0	0	C	none; relationship	fixed BMI curve
observation/abg/index calendar day	176922	0	0	0	C	none; relationship	fixed BMI curve
observation/vbg/index calendar day	176922	0	0	0	C	none; relationship	fixed BMI curve
observation/any gas/index calendar day	176922	0	0	0	C	none; relationship	fixed BMI curve
matched/abg/icd	29639	0	0	0	M abg	none; relationship	fixed BMI curve
matched/abg/abg	29639	0	0	0	M abg	none; relationship	fixed BMI curve

model id	n reference	n transformed outside	pct transformed outside	maximum transformed gap	reference id	policy id	method
matched/any gas/icd	52978	0	0	0	M any gas	none; relationship	fixed BMI curve
matched/any gas/any gas	52978	0	0	0	M any gas	none; relationship	fixed BMI curve

Table 26: Curve-grid and transformed support diagnostics — panel 2/2

model id	reference population id	outcome id	ascertainment window	model form
abg/encounter/spline	C abg encounter	abg	encounter	spline
any gas/encounter/spline	C any gas encounter	any gas	encounter	spline
icd/encounter/spline	C icd	icd	encounter	spline
abg/index calendar day/spline	C abg index calendar day	abg	index calendar day	spline
any gas/index calendar day/spline	C any gas index calendar day	any gas	index calendar day	spline
matched/abg/icd	M abg	icd	encounter	spline
matched/abg/abg	M abg	abg	encounter	spline
matched/any gas/icd	M any gas	icd	encounter	spline
matched/any gas/any gas	M any gas	any gas	encounter	spline
abg/encounter/spline	C abg encounter	abg	encounter	spline
abg/encounter/linear	C abg encounter	abg	encounter	linear
any gas/encounter/spline	C any gas encounter	any gas	encounter	spline
any gas/encounter/linear	C any gas encounter	any gas	encounter	linear
icd/encounter/spline	C icd	icd	encounter	spline
icd/encounter/linear	C icd	icd	encounter	linear
abg/index calendar day/spline	C abg index calendar day	abg	index calendar day	spline
abg/index calendar day/linear	C abg index calendar day	abg	index calendar day	linear
any gas/index calendar day/spline	C any gas index calendar day	any gas	index calendar day	spline
any gas/index calendar day/linear	C any gas index calendar day	any gas	index calendar day	linear
observation/abg/encounter	C	abg	encounter	spline
observation/vbg/encounter	C	vbg	encounter	spline
observation/any gas/encounter	C	any gas	encounter	spline
observation/abg/index calendar day	C	abg	index calendar day	spline
observation/vbg/index calendar day	C	vbg	index calendar day	spline
observation/any gas/index calendar day	C	any gas	index calendar day	spline
matched/abg/icd	M abg	icd	encounter	spline
matched/abg/abg	M abg	abg	encounter	spline
matched/any gas/icd	M any gas	icd	encounter	spline
matched/any gas/any gas	M any gas	any gas	encounter	spline

Table 27: Fixed-index timing transitions

outcome id	primary	day	n	analysis population id
abg	available negative	available negative	8228	primary
abg	positive	available negative	1088	primary
abg	unknown	available negative	0	primary
abg	available negative	positive	0	primary
abg	positive	positive	8000	primary
abg	unknown	positive	0	primary

outcome id	primary	day	n	analysis population id
abg	available negative	unknown	5870	primary
abg	positive	unknown	6453	primary
abg	unknown	unknown	147283	primary
any gas	available negative	available negative	21991	primary
any gas	positive	available negative	1662	primary
any gas	unknown	available negative	0	primary
any gas	available negative	positive	0	primary
any gas	positive	positive	15224	primary
any gas	unknown	positive	0	primary
any gas	available negative	unknown	7366	primary
any gas	positive	unknown	6735	primary
any gas	unknown	unknown	123944	primary

status: assessed; reason: same-row nested comparison; no model selection; df: 2; analysis population id: primary.

Table 28: Same-sample linear checks

outcome id	window	n model	likelihood ratio	p value	ascertainment window
abg	encounter	29639	1.943488	0.3784225	encounter
any gas	encounter	52978	6.050963	0.04853443	encounter
icd	encounter	176922	15.70311	3.891455e-04	encounter
abg	index calendar day	17316	0.1101993	0.9463908	index calendar day
any gas	index calendar day	38877	8.900784	0.01167399	index calendar day

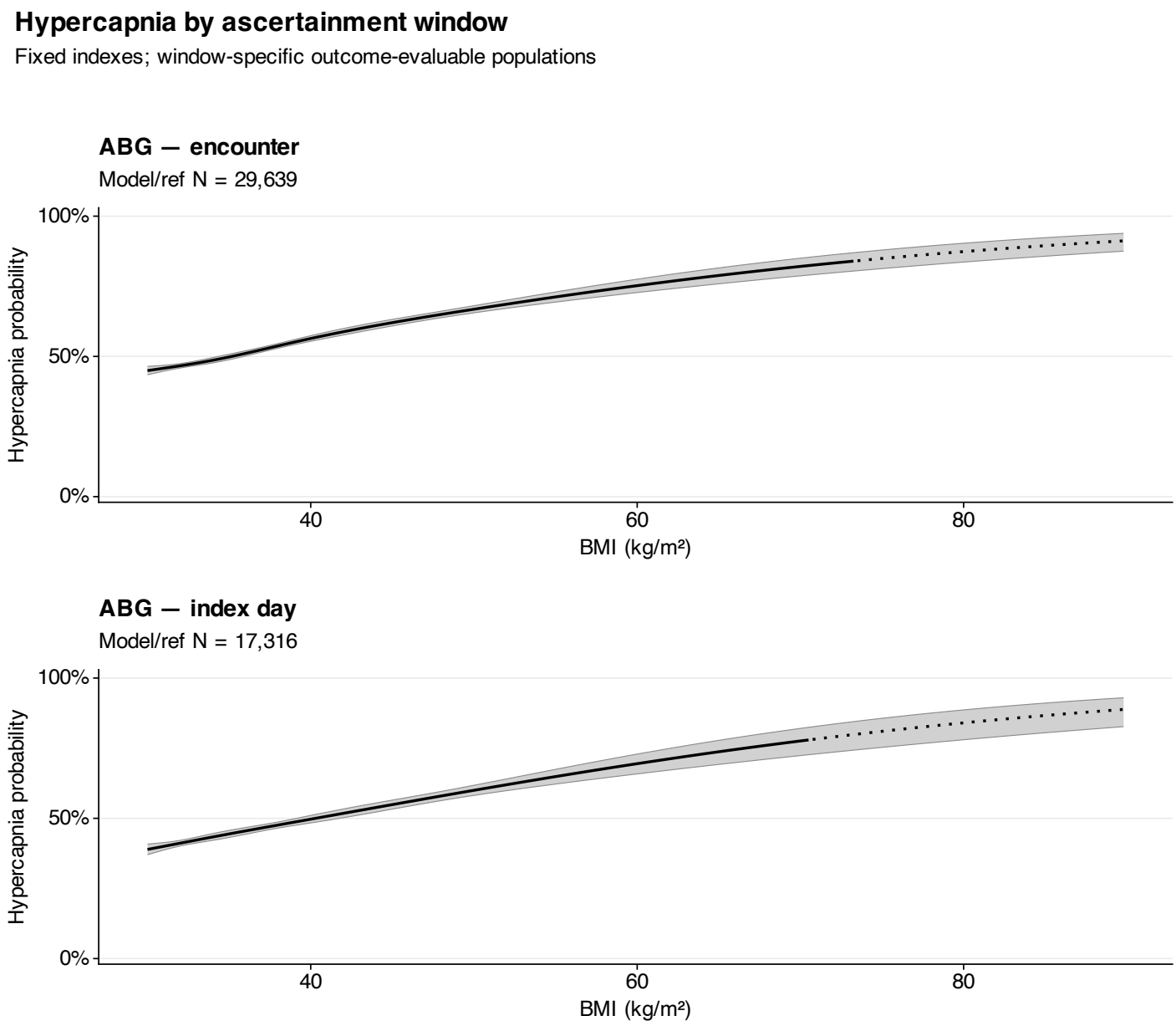


Figure 6: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

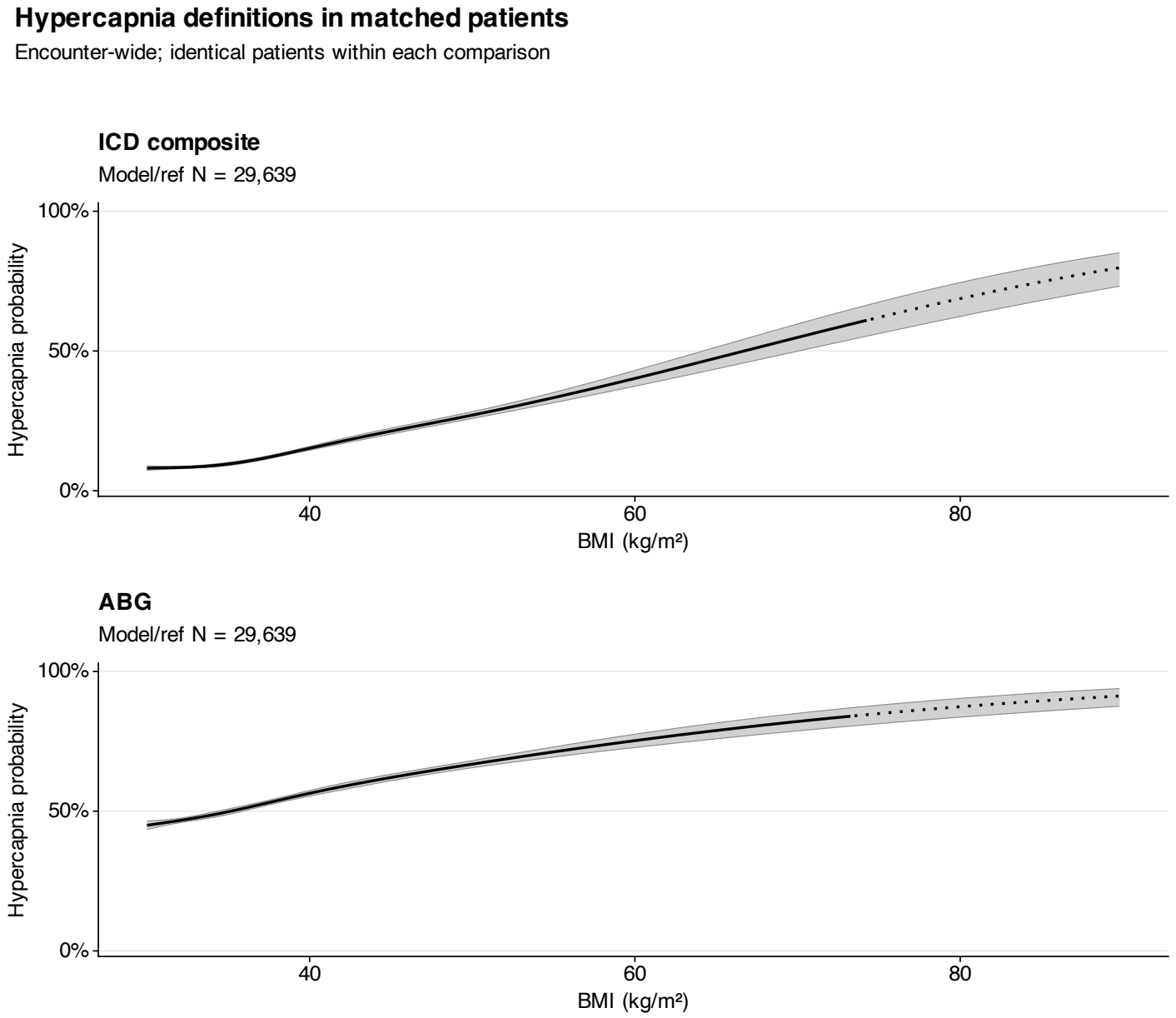


Figure 7: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

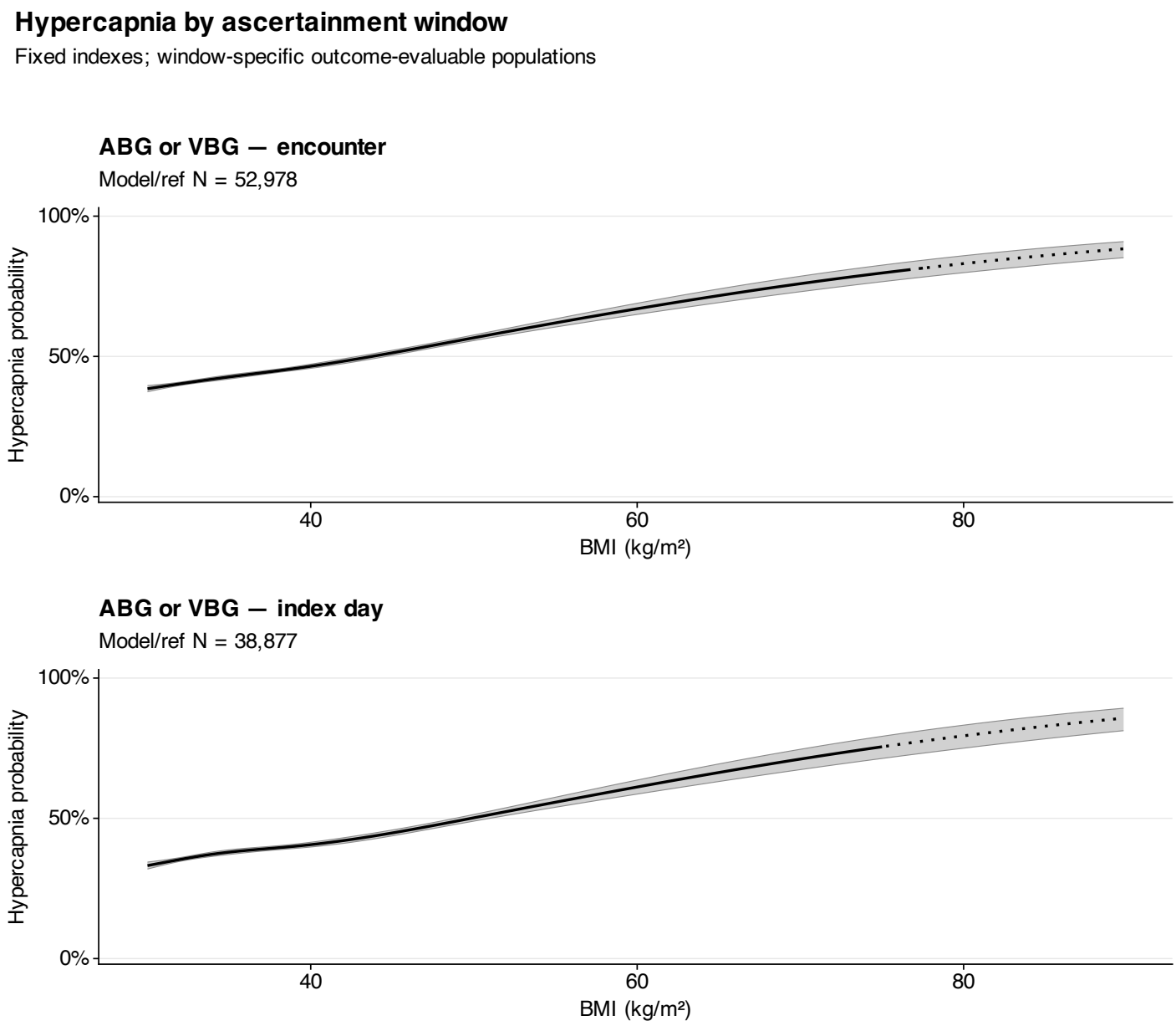


Figure 8: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Encounter versus encounter-start-calendar-day curves reuse the same indexes and fixed spline basis, on shared BMI/probability scales. Each curve standardizes over its displayed model/reference population. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. Pending source or failed models remain explicit. Index day is a calendar-date restriction, not elapsed 24 hours.

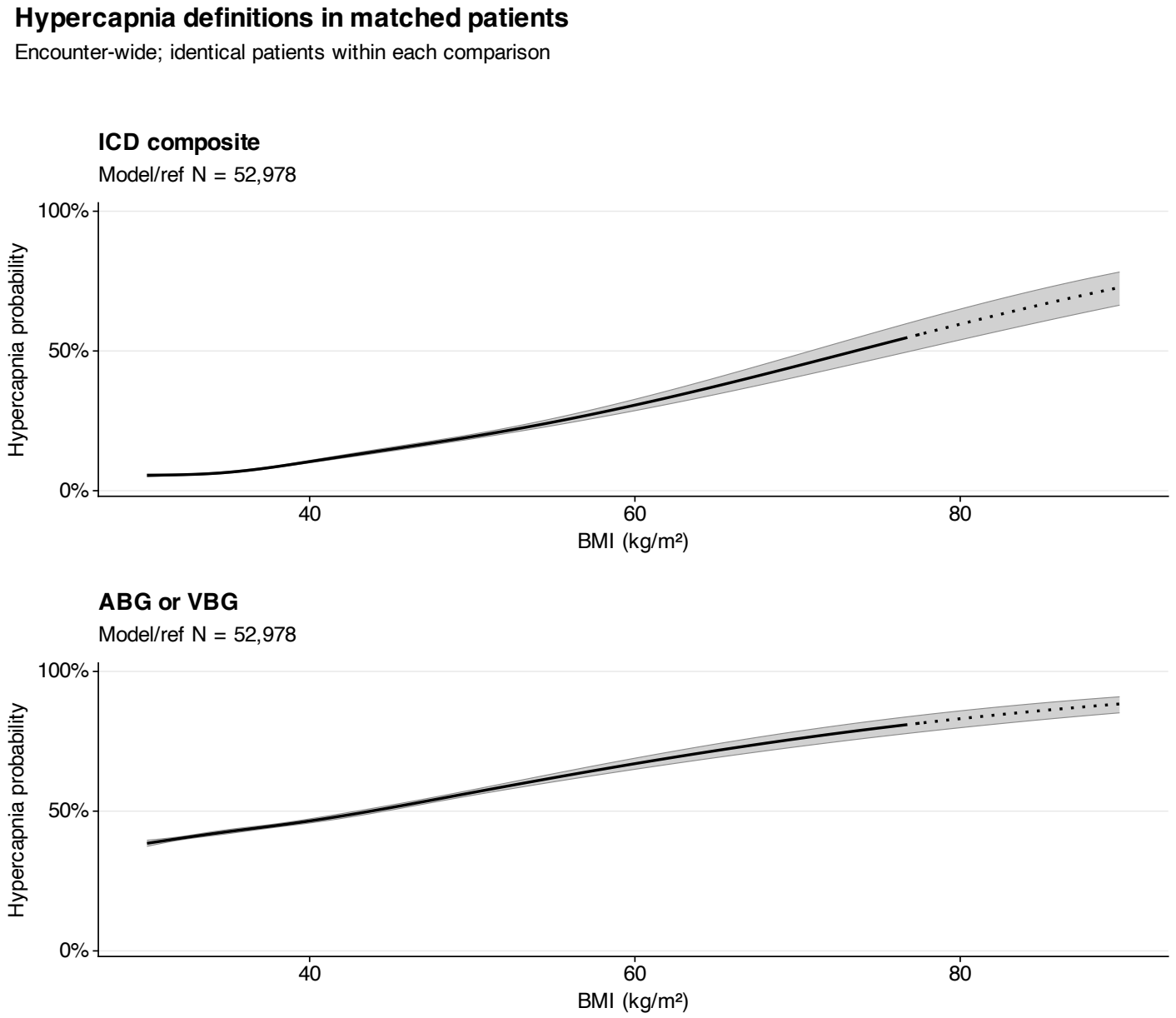


Figure 9: Source-entry population: primary. Subgroup accounting begins with fixed indexes; source/input nodes in these subgroup views do not repeat upstream selection. F5. Matched ICD/gas curves use identical training/reference patients within each pair and shared BMI/probability scales. Arterial and combined pairs can have different membership. Bands are pointwise model-based 95% CIs; dotted segments are locally sparse and dashed segments exceed training BMI support. ICD is the five-code composite including E66.2. Separate bands are not a test of between-model differences.

3.3 Measurement provenance and limited current-measurement checks

stage: Q.

Table 29: Primary BMI sources and lags at fixed indexes — panel 1/2

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	calculated	40to50	negative	bmi lag days	2134	0	2134
non obesity only	calculated	40to50	negative	height lag days	2134	2134	0
non obesity only	calculated	40to50	negative	weight lag days	2134	2134	0
non obesity only	calculated	40to50	negative	component gap days	2134	2134	0
non obesity only	calculated	40to50	positive	bmi lag days	1967	0	1967
non obesity only	calculated	40to50	positive	height lag days	1967	1967	0
non obesity only	calculated	40to50	positive	weight lag days	1967	1967	0
non obesity only	calculated	40to50	positive	component gap days	1967	1967	0
non obesity only	calculated	40to50	unavailable	bmi lag days	6299	0	6299
non obesity only	calculated	40to50	unavailable	height lag days	6299	6299	0
non obesity only	calculated	40to50	unavailable	weight lag days	6299	6299	0
non obesity only	calculated	40to50	unavailable	component gap days	6299	6299	0
non obesity only	calculated	50plus	negative	bmi lag days	697	0	697
non obesity only	calculated	50plus	negative	height lag days	697	697	0
non obesity only	calculated	50plus	negative	weight lag days	697	697	0
non obesity only	calculated	50plus	negative	component gap days	697	697	0
non obesity only	calculated	50plus	positive	bmi lag days	1101	0	1101
non obesity only	calculated	50plus	positive	height lag days	1101	1101	0
non obesity only	calculated	50plus	positive	weight lag days	1101	1101	0
non obesity only	calculated	50plus	positive	component gap days	1101	1101	0
non obesity only	calculated	50plus	unavailable	bmi lag days	3576	0	3576
non obesity only	calculated	50plus	unavailable	height lag days	3576	3576	0
non obesity only	calculated	50plus	unavailable	weight lag days	3576	3576	0
non obesity only	calculated	50plus	unavailable	component gap days	3576	3576	0
non obesity only	calculated	below40	negative	bmi lag days	10302	0	10302
non obesity only	calculated	below40	negative	height lag days	10302	10302	0
non obesity only	calculated	below40	negative	weight lag days	10302	10302	0
non obesity only	calculated	below40	negative	component gap days	10302	10302	0
non obesity only	calculated	below40	positive	bmi lag days	7814	0	7814
non obesity only	calculated	below40	positive	height lag days	7814	7814	0
non obesity only	calculated	below40	positive	weight lag days	7814	7814	0
non obesity only	calculated	below40	positive	component gap days	7814	7814	0

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	calculated	below40	unavailable	bmi lag days	21970	0	21970
non obesity only	calculated	below40	unavailable	height lag days	21970	21970	0
non obesity only	calculated	below40	unavailable	weight lag days	21970	21970	0
non obesity only	calculated	below40	unavailable	component gap days	21970	21970	0
non obesity only	current	40to50	negative	bmi lag days	780	780	0
non obesity only	current	40to50	negative	height lag days	780	761	19
non obesity only	current	40to50	negative	weight lag days	780	769	11
non obesity only	current	40to50	negative	component gap days	780	752	28
non obesity only	current	40to50	positive	bmi lag days	767	767	0
non obesity only	current	40to50	positive	height lag days	767	739	28
non obesity only	current	40to50	positive	weight lag days	767	759	8
non obesity only	current	40to50	positive	component gap days	767	732	35
non obesity only	current	40to50	unavailable	bmi lag days	6224	6224	0
non obesity only	current	40to50	unavailable	height lag days	6224	6131	93
non obesity only	current	40to50	unavailable	weight lag days	6224	6141	83
non obesity only	current	40to50	unavailable	component gap days	6224	6055	169
non obesity only	current	50plus	negative	bmi lag days	4	4	0
non obesity only	current	50plus	negative	height lag days	4	4	0
non obesity only	current	50plus	negative	weight lag days	4	4	0
non obesity only	current	50plus	negative	component gap days	4	4	0
non obesity only	current	50plus	positive	bmi lag days	3	3	0
non obesity only	current	50plus	positive	height lag days	3	3	0
non obesity only	current	50plus	positive	weight lag days	3	3	0
non obesity only	current	50plus	positive	component gap days	3	3	0
non obesity only	current	50plus	unavailable	bmi lag days	19	19	0
non obesity only	current	50plus	unavailable	height lag days	19	18	1
non obesity only	current	50plus	unavailable	weight lag days	19	19	0
non obesity only	current	50plus	unavailable	component gap days	19	18	1
non obesity only	current	below40	negative	bmi lag days	8104	8104	0
non obesity only	current	below40	negative	height lag days	8104	7685	419
non obesity only	current	below40	negative	weight lag days	8104	7861	243
non obesity only	current	below40	negative	component gap days	8104	7459	645
non obesity only	current	below40	positive	bmi lag days	6211	6211	0
non obesity only	current	below40	positive	height lag days	6211	5952	259
non obesity only	current	below40	positive	weight lag days	6211	6073	138

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	current	below40	positive	component gap days	6211	5825	386
non obesity only	current	below40	unavailable	bmi lag days	38326	38326	0
non obesity only	current	below40	unavailable	height lag days	38326	36873	1453
non obesity only	current	below40	unavailable	weight lag days	38326	37547	779
non obesity only	current	below40	unavailable	component gap days	38326	36147	2179
non obesity only	prior	40to50	negative	bmi lag days	876	876	0
non obesity only	prior	40to50	negative	height lag days	876	846	30
non obesity only	prior	40to50	negative	weight lag days	876	864	12
non obesity only	prior	40to50	negative	component gap days	876	839	37
non obesity only	prior	40to50	positive	bmi lag days	728	728	0
non obesity only	prior	40to50	positive	height lag days	728	713	15
non obesity only	prior	40to50	positive	weight lag days	728	721	7
non obesity only	prior	40to50	positive	component gap days	728	708	20
non obesity only	prior	40to50	unavailable	bmi lag days	8084	8084	0
non obesity only	prior	40to50	unavailable	height lag days	8084	7501	583
non obesity only	prior	40to50	unavailable	weight lag days	8084	7994	90
non obesity only	prior	40to50	unavailable	component gap days	8084	7445	639
non obesity only	prior	50plus	negative	bmi lag days	12	12	0
non obesity only	prior	50plus	negative	height lag days	12	12	0
non obesity only	prior	50plus	negative	weight lag days	12	12	0
non obesity only	prior	50plus	negative	component gap days	12	12	0
non obesity only	prior	50plus	positive	bmi lag days	15	15	0
non obesity only	prior	50plus	positive	height lag days	15	15	0
non obesity only	prior	50plus	positive	weight lag days	15	15	0
non obesity only	prior	50plus	positive	component gap days	15	15	0
non obesity only	prior	50plus	unavailable	bmi lag days	97	97	0
non obesity only	prior	50plus	unavailable	height lag days	97	96	1
non obesity only	prior	50plus	unavailable	weight lag days	97	92	5
non obesity only	prior	50plus	unavailable	component gap days	97	92	5
non obesity only	prior	below40	negative	bmi lag days	3531	3531	0
non obesity only	prior	below40	negative	height lag days	3531	3438	93
non obesity only	prior	below40	negative	weight lag days	3531	3474	57
non obesity only	prior	below40	negative	component gap days	3531	3404	127
non obesity only	prior	below40	positive	bmi lag days	1763	1763	0
non obesity only	prior	below40	positive	height lag days	1763	1718	45

subgroup	method	bmi band	gas state	metric	n	known	unknown
non obesity only	prior	below40	positive	weight lag days	1763	1746	17
non obesity only	prior	below40	positive	component gap days	1763	1708	55
non obesity only	prior	below40	unavailable	bmi lag days	16089	16089	0
non obesity only	prior	below40	unavailable	height lag days	16089	14399	1690
non obesity only	prior	below40	unavailable	weight lag days	16089	15842	247
non obesity only	prior	below40	unavailable	component gap days	16089	14219	1870
overlap	calculated	40to50	negative	bmi lag days	472	0	472
overlap	calculated	40to50	negative	height lag days	472	472	0
overlap	calculated	40to50	negative	weight lag days	472	472	0
overlap	calculated	40to50	negative	component gap days	472	472	0
overlap	calculated	40to50	positive	bmi lag days	572	0	572
overlap	calculated	40to50	positive	height lag days	572	572	0
overlap	calculated	40to50	positive	weight lag days	572	572	0
overlap	calculated	40to50	positive	component gap days	572	572	0
overlap	calculated	40to50	unavailable	bmi lag days	1956	0	1956
overlap	calculated	40to50	unavailable	height lag days	1956	1956	0
overlap	calculated	40to50	unavailable	weight lag days	1956	1956	0
overlap	calculated	40to50	unavailable	component gap days	1956	1956	0
overlap	calculated	50plus	negative	bmi lag days	212	0	212
overlap	calculated	50plus	negative	height lag days	212	212	0
overlap	calculated	50plus	negative	weight lag days	212	212	0
overlap	calculated	50plus	negative	component gap days	212	212	0
overlap	calculated	50plus	positive	bmi lag days	384	0	384
overlap	calculated	50plus	positive	height lag days	384	384	0
overlap	calculated	50plus	positive	weight lag days	384	384	0
overlap	calculated	50plus	positive	component gap days	384	384	0
overlap	calculated	50plus	unavailable	bmi lag days	1666	0	1666
overlap	calculated	50plus	unavailable	height lag days	1666	1666	0
overlap	calculated	50plus	unavailable	weight lag days	1666	1666	0
overlap	calculated	50plus	unavailable	component gap days	1666	1666	0
overlap	calculated	below40	negative	bmi lag days	341	0	341
overlap	calculated	below40	negative	height lag days	341	341	0
overlap	calculated	below40	negative	weight lag days	341	341	0
overlap	calculated	below40	negative	component gap days	341	341	0
overlap	calculated	below40	positive	bmi lag days	390	0	390

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	calculated	below40	positive	height lag days	390	390	0
overlap	calculated	below40	positive	weight lag days	390	390	0
overlap	calculated	below40	positive	component gap days	390	390	0
overlap	calculated	below40	unavailable	bmi lag days	1167	0	1167
overlap	calculated	below40	unavailable	height lag days	1167	1167	0
overlap	calculated	below40	unavailable	weight lag days	1167	1167	0
overlap	calculated	below40	unavailable	component gap days	1167	1167	0
overlap	current	40to50	negative	bmi lag days	1284	1284	0
overlap	current	40to50	negative	height lag days	1284	1201	83
overlap	current	40to50	negative	weight lag days	1284	1235	49
overlap	current	40to50	negative	component gap days	1284	1154	130
overlap	current	40to50	positive	bmi lag days	1203	1203	0
overlap	current	40to50	positive	height lag days	1203	1148	55
overlap	current	40to50	positive	weight lag days	1203	1176	27
overlap	current	40to50	positive	component gap days	1203	1124	79
overlap	current	40to50	unavailable	bmi lag days	13550	13550	0
overlap	current	40to50	unavailable	height lag days	13550	12771	779
overlap	current	40to50	unavailable	weight lag days	13550	13235	315
overlap	current	40to50	unavailable	component gap days	13550	12488	1062
overlap	current	50plus	negative	bmi lag days	6	6	0
overlap	current	50plus	negative	height lag days	6	6	0
overlap	current	50plus	negative	weight lag days	6	5	1
overlap	current	50plus	negative	component gap days	6	5	1
overlap	current	50plus	positive	bmi lag days	2	2	0
overlap	current	50plus	positive	height lag days	2	2	0
overlap	current	50plus	positive	weight lag days	2	2	0
overlap	current	50plus	positive	component gap days	2	2	0
overlap	current	50plus	unavailable	bmi lag days	49	49	0
overlap	current	50plus	unavailable	height lag days	49	47	2
overlap	current	50plus	unavailable	weight lag days	49	47	2
overlap	current	50plus	unavailable	component gap days	49	45	4
overlap	current	below40	negative	bmi lag days	330	330	0
overlap	current	below40	negative	height lag days	330	323	7
overlap	current	below40	negative	weight lag days	330	306	24
overlap	current	below40	negative	component gap days	330	300	30

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	current	below40	positive	bmi lag days	434	434	0
overlap	current	below40	positive	height lag days	434	426	8
overlap	current	below40	positive	weight lag days	434	402	32
overlap	current	below40	positive	component gap days	434	396	38
overlap	current	below40	unavailable	bmi lag days	1960	1960	0
overlap	current	below40	unavailable	height lag days	1960	1935	25
overlap	current	below40	unavailable	weight lag days	1960	1914	46
overlap	current	below40	unavailable	component gap days	1960	1891	69
overlap	prior	40to50	negative	bmi lag days	152	152	0
overlap	prior	40to50	negative	height lag days	152	149	3
overlap	prior	40to50	negative	weight lag days	152	152	0
overlap	prior	40to50	negative	component gap days	152	149	3
overlap	prior	40to50	positive	bmi lag days	166	166	0
overlap	prior	40to50	positive	height lag days	166	166	0
overlap	prior	40to50	positive	weight lag days	166	164	2
overlap	prior	40to50	positive	component gap days	166	164	2
overlap	prior	40to50	unavailable	bmi lag days	1723	1723	0
overlap	prior	40to50	unavailable	height lag days	1723	1624	99
overlap	prior	40to50	unavailable	weight lag days	1723	1718	5
overlap	prior	40to50	unavailable	component gap days	1723	1623	100
overlap	prior	50plus	negative	bmi lag days	3	3	0
overlap	prior	50plus	negative	height lag days	3	3	0
overlap	prior	50plus	negative	weight lag days	3	3	0
overlap	prior	50plus	negative	component gap days	3	3	0
overlap	prior	50plus	positive	bmi lag days	1	1	0
overlap	prior	50plus	positive	height lag days	1	1	0
overlap	prior	50plus	positive	weight lag days	1	1	0
overlap	prior	50plus	positive	component gap days	1	1	0
overlap	prior	50plus	unavailable	bmi lag days	28	28	0
overlap	prior	50plus	unavailable	height lag days	28	28	0
overlap	prior	50plus	unavailable	weight lag days	28	28	0
overlap	prior	50plus	unavailable	component gap days	28	28	0
overlap	prior	below40	negative	bmi lag days	117	117	0
overlap	prior	below40	negative	height lag days	117	117	0
overlap	prior	below40	negative	weight lag days	117	113	4

subgroup	method	bmi band	gas state	metric	n	known	unknown
overlap	prior	below40	negative	component gap days	117	113	4
overlap	prior	below40	positive	bmi lag days	100	100	0
overlap	prior	below40	positive	height lag days	100	99	1
overlap	prior	below40	positive	weight lag days	100	99	1
overlap	prior	below40	positive	component gap days	100	98	2
overlap	prior	below40	unavailable	bmi lag days	1161	1161	0
overlap	prior	below40	unavailable	height lag days	1161	1026	135
overlap	prior	below40	unavailable	weight lag days	1161	1141	20
overlap	prior	below40	unavailable	component gap days	1161	1014	147

Table 30: Primary BMI sources and lags at fixed indexes — panel 2/2

subgroup	median	maximum	status
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6679	assessed
non obesity only	0	6473	assessed
non obesity only	0	6574	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	3457	assessed
non obesity only	0	5901	assessed
non obesity only	0	5901	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	7545	assessed
non obesity only	0	7211	assessed
non obesity only	0	7325	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	4164	assessed
non obesity only	0	6270	assessed
non obesity only	0	6255	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6641	assessed
non obesity only	0	6641	assessed
non obesity only	0	5160	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6817	assessed
non obesity only	0	7830	assessed
non obesity only	0	7830	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	7064	assessed
non obesity only	0	6901	assessed
non obesity only	0	7064	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	6433	assessed
non obesity only	0	7318	assessed
non obesity only	0	6593	assessed
non obesity only	unavailable	unavailable	unavailable or unknown
non obesity only	0	8039	assessed
non obesity only	0	6935	assessed
non obesity only	0	8039	assessed
non obesity only	0	0	assessed
non obesity only	0	2717	assessed
non obesity only	0	4165	assessed
non obesity only	0	4170	assessed
non obesity only	0	0	assessed
non obesity only	0	2646	assessed
non obesity only	0	3567	assessed
non obesity only	0	3569	assessed
non obesity only	0	0	assessed
non obesity only	0	6469	assessed

subgroup	median	maximum	status
non obesity only	0	4192	assessed
non obesity only	0	6469	assessed
non obesity only	-5	0	assessed
non obesity only	0	1749	assessed
non obesity only	0	0	assessed
non obesity only	4.5	1749	assessed
non obesity only	0	0	assessed
non obesity only	0	0	assessed
non obesity only	0	0	assessed
non obesity only	0	2	assessed
non obesity only	0	0	assessed
non obesity only	0	2	assessed
non obesity only	0	409	assessed
non obesity only	0	409	assessed
non obesity only	0	0	assessed
non obesity only	0	6685	assessed
non obesity only	0	4250	assessed
non obesity only	0	6685	assessed
non obesity only	0	0	assessed
non obesity only	0	6658	assessed
non obesity only	0	4372	assessed
non obesity only	0	6658	assessed
non obesity only	0	0	assessed
non obesity only	0	6984	assessed
non obesity only	0	4399	assessed
non obesity only	0	6984	assessed
non obesity only	190.5	6240	assessed
non obesity only	16	6240	assessed
non obesity only	3	5376	assessed
non obesity only	6	6119	assessed
non obesity only	186.5	6524	assessed
non obesity only	4	6641	assessed
non obesity only	1	4964	assessed
non obesity only	0	6482	assessed
non obesity only	141	6888	assessed
non obesity only	29	6756	assessed
non obesity only	16	6609	assessed
non obesity only	0	6756	assessed
non obesity only	201.5	2304	assessed
non obesity only	0	467	assessed
non obesity only	0	349	assessed
non obesity only	0	429	assessed
non obesity only	382	4581	assessed
non obesity only	1	1072	assessed
non obesity only	1	1072	assessed
non obesity only	0	326	assessed
non obesity only	357	3937	assessed
non obesity only	0	826	assessed
non obesity only	0	2534	assessed
non obesity only	0	2534	assessed
non obesity only	132	5223	assessed
non obesity only	21	5223	assessed
non obesity only	6	6669	assessed
non obesity only	15	5515	assessed
non obesity only	93	7231	assessed
non obesity only	11	5934	assessed
non obesity only	10	5934	assessed
non obesity only	1	4677	assessed
non obesity only	105	7729	assessed
non obesity only	62	7218	assessed
non obesity only	38	7729	assessed
non obesity only	0	6069	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4039	assessed
overlap	0	5859	assessed
overlap	0	5859	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4056	assessed
overlap	0	5712	assessed
overlap	0	5712	assessed
overlap	unavailable	unavailable	unavailable or unknown

subgroup	median	maximum	status
overlap	0	5701	assessed
overlap	0	6495	assessed
overlap	0	6441	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	1761	assessed
overlap	0	5560	assessed
overlap	0	5560	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	2286	assessed
overlap	0	5599	assessed
overlap	0	5599	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4505	assessed
overlap	0	5605	assessed
overlap	0	5605	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4404	assessed
overlap	0	5871	assessed
overlap	0	5871	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	3386	assessed
overlap	0	5687	assessed
overlap	2	5687	assessed
overlap	unavailable	unavailable	unavailable or unknown
overlap	0	4144	assessed
overlap	0	5902	assessed
overlap	0	5903	assessed
overlap	0	0	assessed
overlap	0	6618	assessed
overlap	0	3739	assessed
overlap	0	6618	assessed
overlap	0	0	assessed
overlap	0	4569	assessed
overlap	0	3154	assessed
overlap	0	4569	assessed
overlap	0	0	assessed
overlap	0	6802	assessed
overlap	0	4434	assessed
overlap	0	6802	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	2322	assessed
overlap	0	0	assessed
overlap	0	2322	assessed
overlap	0	0	assessed
overlap	0	2798	assessed
overlap	0	3909	assessed
overlap	0	3909	assessed
overlap	0	0	assessed
overlap	0	2750	assessed
overlap	0	3844	assessed
overlap	0	3844	assessed
overlap	0	0	assessed
overlap	0	6976	assessed
overlap	0	3962	assessed
overlap	0	6976	assessed
overlap	533.5	4797	assessed
overlap	0	3649	assessed
overlap	168.5	3477	assessed
overlap	163	3289	assessed
overlap	583	5694	assessed
overlap	0	3227	assessed
overlap	0	3595	assessed
overlap	2	3595	assessed

subgroup	median	maximum	status
overlap	489	7039	assessed
overlap	59	6769	assessed
overlap	100	6455	assessed
overlap	0	4835	assessed
overlap	149	681	assessed
overlap	0	149	assessed
overlap	75	149	assessed
overlap	1	75	assessed
overlap	190	190	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	0	0	assessed
overlap	1060	4159	assessed
overlap	0	2421	assessed
overlap	1	1648	assessed
overlap	0	1659	assessed
overlap	303	4649	assessed
overlap	0	3485	assessed
overlap	148	3360	assessed
overlap	84	3360	assessed
overlap	325	3698	assessed
overlap	0	3698	assessed
overlap	134	3698	assessed
overlap	39	2303	assessed
overlap	135	6981	assessed
overlap	50.5	6981	assessed
overlap	63	6172	assessed
overlap	0	4425	assessed

3.3.1 primary_current

family: supporting; window: encounter; model form: spline; source status: qualified; formula text: .outcome ~ bmi_ns1 + bmi_ns2 + bmi_ns3 + age_at_encounter + sex + race_ethnicity + location; basis id: 026b7a1829cd49840558b84e972374c0e1cb1f3a2f2304811582736b3d37f105; weight specification: none; covariance method: model-based binomial; empirical reference fixed; n eligible: 126480; n excluded adjustment: 0; n parameters: 15; model rank: 15; status: successful; failure reason: ; warnings: ; analysis population id: primary_current; ascertainment window: encounter; population id: primary_current; source reason: unavailable.

Table 31: Current-measurement fit status — panel 1/2

model id	outcome id	training population id	field	n excluded outcome	n model	n patients	n events
abg/encounter/spline	abg	C abg encounter	abg	103429	23051	23051	12201
any gas/encounter/spline	any gas	C any gas encounter	any gas	87865	38615	38615	17769
icd/encounter/spline	icd	C icd	icd	0	126480	126480	6258

Table 32: Current-measurement fit status — panel 2/2

model id	n nonevents	fit runtime
abg/encounter/spline	10850	0.098
any gas/encounter/spline	20846	0.161
icd/encounter/spline	120222	0.56

Maximum decrement: 1.

Table 33: Current-measurement scenario — panel 1/2

Model	Reference	Model N	Reference N	Events	Patients shifted	Mean decrement	Eligible (%)
abg/encounter/spline	C abg encounter	23051	23051	12201	22714	0.9853802	98.54
any gas/encounter/ spline	C any gas encounter	38615	38615	17769	38053	0.9854461	98.54
icd/encounter/spline	C icd	126480	126480	6258	124789	0.9866303	98.66

Table 34: Current-measurement scenario — panel 2/2

Model	Observed probability % (95% CI)	Shifted probability % (95% CI)	Difference per 1000 (95% CI)	PIF % (95% CI)
abg/encounter/spline	52.93 (52.29, 53.57)	51.85 (51.17, 52.53)	10.77 (8.423, 13.12)	2.035 (1.589, 2.478)
any gas/encounter/ spline	46.02 (45.53, 46.51)	45.12 (44.61, 45.64)	8.917 (7.182, 10.65)	1.938 (1.56, 2.315)
icd/encounter/spline	4.948 (4.831, 5.067)	4.688 (4.572, 4.806)	2.6 (2.263, 2.936)	5.254 (4.582, 5.922)

3.4 Same-day arterial PCO_2/pH co-occurrence

This secondary analysis uses any recorded calendar date within the existing index encounter with both usable ordinary arterial PCO_2 and arterial pH. It is **same recorded-day co-occurrence; not verified same-sample pairing**. The export dates when results were recorded; it does not identify a shared blood sample, collection time or elapsed 24-hour interval. Validity screens are $5<PCO_2<200$ mm Hg and $6.5<pH<7.8$, applied before display rounding.

The source endpoints use the same patients with at least one co-observed day: any $PaCO_2 \geq 45$ on such a day, and any such day also containing pH <7.35. Results may come from different samples. Elevated PCO_2 on one date and low pH on another cannot establish co-occurrence. Patients without a valid co-observed day have unknown supplemental outcomes. Additional unusable or single-analyte records remain audited without erasing usable days. A negative describes eligible recorded evidence only. The primary numeric BMI spline basis, inherited adjustment and common supplemental reference population are retained. A single unpenalized multinomial logistic spline assigns each patient to no documented hypercapnia, hypercapnia without documented same-day acidemia, or hypercapnia with same-day acidemia. Any eligible hypercapnia–acidemia day takes priority, including patients with other hypercapnic days without acidemia. The baseline outcome is no documented hypercapnia. All three probabilities are standardized over the same patients; full cross-category covariance supplies pointwise intervals and comparisons of component changes. No additional PIF is calculated.

The full observed BMI range is the main relationship, with sparse regions flagged. BMI 40–45 is a prespecified numerical summary, not a biological cutoff or fitting restriction. Every supported five-unit interval is also reported. These are descriptive pointwise contrasts without multiplicity adjustment; separate significance counts do not establish differences in associations.

Table 35: Recorded-day source qualification

module	status	reason	n primary complete	n coobserved	analysis population id
same day arterial cooccurrence v1	qualified	Recorded-day evidence assessed; zeros refer only to co-observed days	176922	22195	primary

Table 36: Four-stage patient/evidence reconciliation; raw proportions with Wilson 95% intervals

Stage	Patients	Events	Observed	Status
Original arterial-evaluable	29639	15541	52.43% (95% CI 51.87–53.00%)	assessed
Co-observed patients: original arterial outcome	22195	11690	52.67% (95% CI 52.01–53.33%)	qualified
Same patients: hypercapnia on co-observed days	22195	11669	52.57% (95% CI 51.92–53.23%)	qualified
Same patients: same-day hypercapnia and acidemia	22195	8809	39.69% (95% CI 39.05–40.33%)	qualified

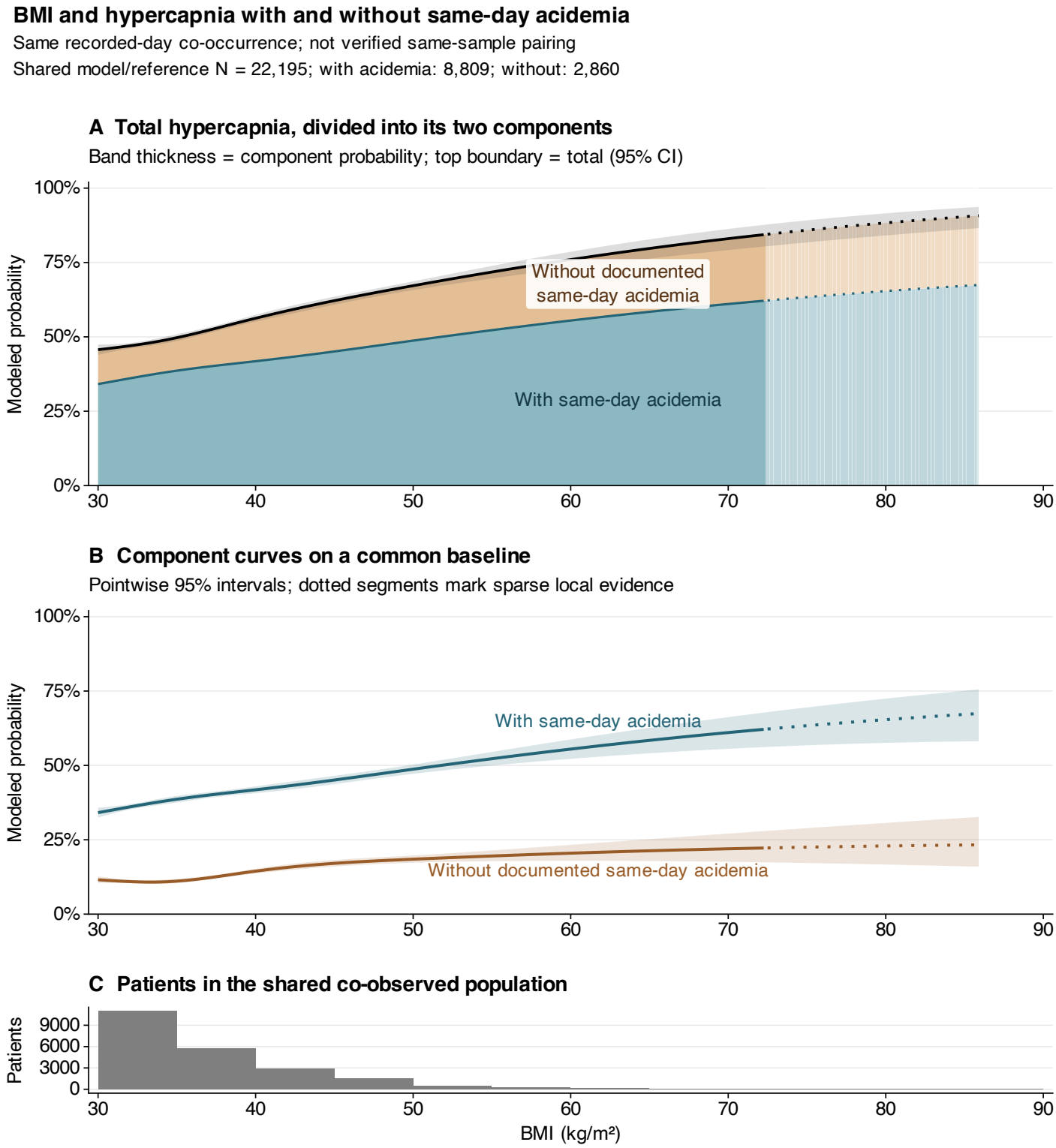


Figure 10: Same recorded-day co-occurrence; not verified same-sample pairing. Primary fixed-index patients with both arterial analytes share inherited adjustment and the primary BMI spline. Any eligible day with $PaCO_2$ ≥45 mm Hg and pH <7.35 assigns the with-acidemia category; remaining hypercapnic patients have no documented same-day acidemia. Different samples may contribute. A: band thicknesses give component probabilities; the upper boundary is total hypercapnia. B: components share a baseline. Shading denotes pointwise 95% intervals using full joint covariance. C: shared-population BMI frequencies. Fading/dotted lines mark local windows below 30 patients or five events/non-events; marginal checks do not establish joint support. These recorded outcomes do not establish simultaneous physiology, acute respiratory failure or compensation. Selection, treatment timing and confounding remain limitations.

Table 37: Mutually exclusive patient categories; any acidemia co-occurrence takes priority

category id	n	N	status	definition	analysis population id
no hypercapnia	10526	22195	qualified	No documented hypercapnia on co-observed days	primary
without acidemia	2860	22195	qualified	Hypercapnia without documented same-day acidemia on any eligible day	primary
with acidemia	8809	22195	qualified	Any eligible day with hypercapnia and acidemia; takes priority over other days	primary

Table 38: Component magnitudes across BMI: absolute changes in percentage points and probability ratios, with 95% intervals

BMI interval	Outcome	Lower BMI probability % (95% CI)	Higher BMI probability % (95% CI)	Change pp (95% CI)	Probability ratio (95% CI)	Status
30 to 35	No documented hypercapnia	54.32 (52.58 to 56.06)	50.28 (49.14 to 51.42)	-4.04 (-6.42 to -1.67)	0.93 (0.88 to 0.97)	Estimated
30 to 35	Hypercapnia without documented same-day acidemia	11.55 (10.48 to 12.71)	11.10 (10.41 to 11.82)	-0.45 (-1.95 to 1.04)	0.96 (0.84 to 1.10)	Estimated
30 to 35	Hypercapnia with same-day acidemia	34.13 (32.51 to 35.79)	38.62 (37.53 to 39.73)	4.50 (2.24 to 6.75)	1.13 (1.06 to 1.21)	Estimated
30 to 35	Any hypercapnia	45.68 (43.94 to 47.42)	49.72 (48.58 to 50.86)	4.04 (1.67 to 6.42)	1.09 (1.03 to 1.14)	Estimated
30 to 35	With minus without: difference in changes	Not estimated	Not estimated	4.95 (1.94 to 7.96)	Not estimated	Estimated
35 to 40	No documented hypercapnia	50.28 (49.14 to 51.42)	43.74 (42.58 to 44.91)	-6.54 (-8.05 to -5.02)	0.87 (0.84 to 0.90)	Estimated
35 to 40	Hypercapnia without documented same-day acidemia	11.10 (10.41 to 11.82)	14.45 (13.68 to 15.26)	3.35 (2.40 to 4.30)	1.30 (1.21 to 1.40)	Estimated
35 to 40	Hypercapnia with same-day acidemia	38.62 (37.53 to 39.73)	41.81 (40.67 to 42.95)	3.18 (1.72 to 4.64)	1.08 (1.04 to 1.12)	Estimated
35 to 40	Any hypercapnia	49.72 (48.58 to 50.86)	56.26 (55.09 to 57.42)	6.54 (5.02 to 8.05)	1.13 (1.10 to 1.16)	Estimated
35 to 40	With minus without: difference in changes	Not estimated	Not estimated	-0.17 (-2.12 to 1.77)	Not estimated	Estimated
40 to 45	No documented hypercapnia	43.74 (42.58 to 44.91)	37.83 (36.49 to 39.19)	-5.91 (-6.63 to -5.19)	0.86 (0.85 to 0.88)	Estimated
40 to 45	Hypercapnia without documented same-day acidemia	14.45 (13.68 to 15.26)	17.08 (16.01 to 18.21)	2.63 (2.03 to 3.23)	1.18 (1.14 to 1.22)	Estimated
40 to 45	Hypercapnia with same-day acidemia	41.81 (40.67 to 42.95)	45.09 (43.68 to 46.50)	3.28 (2.54 to 4.02)	1.08 (1.06 to 1.10)	Estimated
40 to 45	Any hypercapnia	56.26 (55.09 to 57.42)	62.17 (60.81 to 63.51)	5.91 (5.19 to 6.63)	1.11 (1.09 to 1.12)	Estimated
40 to 45	With minus without: difference in changes	Not estimated	Not estimated	0.65 (-0.49 to 1.79)	Not estimated	Estimated
45 to 50	No documented hypercapnia	37.83 (36.49 to 39.19)	32.80 (31.34 to 34.29)	-5.03 (-6.11 to -3.96)	0.87 (0.84 to 0.89)	Estimated
45 to 50	Hypercapnia without documented same-day acidemia	17.08 (16.01 to 18.21)	18.47 (17.28 to 19.72)	1.39 (0.60 to 2.18)	1.08 (1.03 to 1.13)	Estimated
45 to 50	Hypercapnia with same-day acidemia	45.09 (43.68 to 46.50)	48.73 (47.17 to 50.29)	3.64 (2.58 to 4.71)	1.08 (1.06 to 1.11)	Estimated
45 to 50	Any hypercapnia	62.17 (60.81 to 63.51)	67.20 (65.71 to 68.66)	5.03 (3.96 to 6.11)	1.08 (1.06 to 1.10)	Estimated
45 to 50	With minus without: difference in changes	Not estimated	Not estimated	2.25 (0.72 to 3.79)	Not estimated	Estimated

BMI interval	Outcome	Lower BMI probability % (95% CI)	Higher BMI probability % (95% CI)	Change pp (95% CI)	Probability ratio (95% CI)	Status
50 to 55	No documented hypercapnia	32.80 (31.34 to 34.29)	28.21 (26.15 to 30.36)	-4.59 (-5.61 to -3.58)	0.86 (0.83 to 0.89)	Estimated
50 to 55	Hypercapnia without documented same-day acidemia	18.47 (17.28 to 19.72)	19.55 (17.80 to 21.42)	1.08 (0.12 to 2.04)	1.06 (1.01 to 1.11)	Estimated
50 to 55	Hypercapnia with same-day acidemia	48.73 (47.17 to 50.29)	52.25 (49.95 to 54.53)	3.51 (2.34 to 4.69)	1.07 (1.05 to 1.10)	Estimated
50 to 55	Any hypercapnia	67.20 (65.71 to 68.66)	71.79 (69.64 to 73.85)	4.59 (3.58 to 5.61)	1.07 (1.05 to 1.08)	Estimated
50 to 55	With minus without: difference in changes	Not estimated	Not estimated	2.44 (0.55 to 4.32)	Not estimated	Estimated
55 to 60	No documented hypercapnia	28.21 (26.15 to 30.36)	24.02 (21.38 to 26.88)	-4.18 (-4.99 to -3.38)	0.85 (0.82 to 0.89)	Estimated
55 to 60	Hypercapnia without documented same-day acidemia	19.55 (17.80 to 21.42)	20.49 (17.94 to 23.30)	0.95 (-0.08 to 1.97)	1.05 (1.00 to 1.10)	Estimated
55 to 60	Hypercapnia with same-day acidemia	52.25 (49.95 to 54.53)	55.48 (52.20 to 58.72)	3.24 (2.07 to 4.40)	1.06 (1.04 to 1.08)	Estimated
55 to 60	Any hypercapnia	71.79 (69.64 to 73.85)	75.98 (73.12 to 78.62)	4.18 (3.38 to 4.99)	1.06 (1.05 to 1.07)	Estimated
55 to 60	With minus without: difference in changes	Not estimated	Not estimated	2.29 (0.25 to 4.34)	Not estimated	Estimated
60 to 65	No documented hypercapnia	24.02 (21.38 to 26.88)	20.28 (17.23 to 23.72)	-3.74 (-4.32 to -3.16)	0.84 (0.81 to 0.89)	Estimated
60 to 65	Hypercapnia without documented same-day acidemia	20.49 (17.94 to 23.30)	21.30 (17.85 to 25.21)	0.81 (-0.28 to 1.89)	1.04 (0.99 to 1.09)	Estimated
60 to 65	Hypercapnia with same-day acidemia	55.48 (52.20 to 58.72)	58.42 (54.07 to 62.64)	2.94 (1.77 to 4.10)	1.05 (1.03 to 1.07)	Estimated
60 to 65	Any hypercapnia	75.98 (73.12 to 78.62)	79.72 (76.28 to 82.77)	3.74 (3.16 to 4.32)	1.05 (1.04 to 1.06)	Estimated
60 to 65	With minus without: difference in changes	Not estimated	Not estimated	2.13 (-0.04 to 4.30)	Not estimated	Estimated
65 to 70	No documented hypercapnia	20.28 (17.23 to 23.72)	16.99 (13.72 to 20.84)	-3.29 (-3.67 to -2.92)	0.84 (0.80 to 0.88)	Estimated
65 to 70	Hypercapnia without documented same-day acidemia	21.30 (17.85 to 25.21)	21.97 (17.59 to 27.08)	0.67 (-0.46 to 1.80)	1.03 (0.98 to 1.08)	Estimated
65 to 70	Hypercapnia with same-day acidemia	58.42 (54.07 to 62.64)	61.04 (55.59 to 66.24)	2.62 (1.45 to 3.79)	1.04 (1.03 to 1.06)	Estimated
65 to 70	Any hypercapnia	79.72 (76.28 to 82.77)	83.01 (79.16 to 86.28)	3.29 (2.92 to 3.67)	1.04 (1.04 to 1.04)	Estimated
65 to 70	With minus without: difference in changes	Not estimated	Not estimated	1.95 (-0.31 to 4.22)	Not estimated	Estimated
70 to 75	No documented hypercapnia	16.99 (13.72 to 20.84)	14.13 (10.83 to 18.24)	-2.86 (-3.06 to -2.66)	0.83 (0.79 to 0.88)	Estimated; sparse endpoint
70 to 75	Hypercapnia without documented same-day acidemia	21.97 (17.59 to 27.08)	22.51 (17.18 to 28.91)	0.54 (-0.62 to 1.70)	1.02 (0.98 to 1.07)	Estimated; sparse endpoint
70 to 75	Hypercapnia with same-day acidemia	61.04 (55.59 to 66.24)	63.36 (56.75 to 69.51)	2.32 (1.13 to 3.50)	1.04 (1.02 to 1.05)	Estimated; sparse endpoint
70 to 75	Any hypercapnia	83.01 (79.16 to 86.28)	85.87 (81.76 to 89.17)	2.86 (2.66 to 3.06)	1.03 (1.03 to 1.04)	Estimated; sparse endpoint
70 to 75	With minus without: difference in changes	Not estimated	Not estimated	1.78 (-0.56 to 4.11)	Not estimated	Estimated; sparse endpoint
75 to 80	No documented hypercapnia	14.13 (10.83 to 18.24)	11.69 (8.48 to 15.90)	-2.45 (-2.55 to -2.34)	0.83 (0.78 to 0.87)	Estimated; sparse endpoint
75 to 80	Hypercapnia without documented same-day acidemia	22.51 (17.18 to 28.91)	22.93 (16.67 to 30.67)	0.42 (-0.76 to 1.60)	1.02 (0.97 to 1.07)	Estimated; sparse endpoint
75 to 80	Hypercapnia with same-day acidemia	63.36 (56.75 to 69.51)	65.39 (57.57 to 72.45)	2.03 (0.83 to 3.22)	1.03 (1.02 to 1.05)	Estimated; sparse endpoint

BMI interval	Outcome	Lower BMI probability % (95% CI)	Higher BMI probability % (95% CI)	Change pp (95% CI)	Probability ratio (95% CI)	Status
75 to 80	Any hypercapnia	85.87 (81.76 to 89.17)	88.31 (84.10 to 91.52)	2.45 (2.34 to 2.55)	1.03 (1.03 to 1.03)	Estimated; sparse endpoint
75 to 80	With minus without: difference in changes	Not estimated	Not estimated	1.61 (-0.77 to 3.99)	Not estimated	Estimated; sparse endpoint
80 to 85	No documented hypercapnia	11.69 (8.48 to 15.90)	9.61 (6.60 to 13.81)	-2.07 (-2.22 to -1.93)	0.82 (0.78 to 0.87)	Estimated; sparse endpoint
80 to 85	Hypercapnia without documented same-day acidemia	22.93 (16.67 to 30.67)	23.24 (16.07 to 32.38)	0.31 (-0.88 to 1.51)	1.01 (0.97 to 1.06)	Estimated; sparse endpoint
80 to 85	Hypercapnia with same-day acidemia	65.39 (57.57 to 72.45)	67.15 (58.08 to 75.10)	1.76 (0.55 to 2.97)	1.03 (1.01 to 1.04)	Estimated; sparse endpoint
80 to 85	Any hypercapnia	88.31 (84.10 to 91.52)	90.39 (86.19 to 93.40)	2.07 (1.93 to 2.22)	1.02 (1.02 to 1.03)	Estimated; sparse endpoint
80 to 85	With minus without: difference in changes	Not estimated	Not estimated	1.45 (-0.95 to 3.84)	Not estimated	Estimated; sparse endpoint

Table 39: Joint probability and stacked-component reconciliation

Field	Value
model id	primary::same day abg/encounter/multinomial spline
n grid	225
maximum probability sum error	1.187939e-14
maximum component sum error	1.110223e-16
status	assessed
analysis population id	primary
population id	primary
outcome id	three category
ascertainment window	encounter
model form	multinomial spline

Requiring both analytes on the same date introduces additional selection. The reconciliation separates patient restriction, restriction to co-observed days, and addition of low pH. Values can represent different samples or periods before and after treatment; co-occurrence does not establish simultaneous hypercapnic acidemia, acute respiratory failure or chronic compensation. Primary arterial findings on other dates remain unchanged. The joint model guarantees coherent category probabilities without clipping. In the stacked panel, band thickness is the component probability and the upper boundary is total hypercapnia. The common-baseline panel shows both components with their own intervals. The signed with-minus-without change contrasts use joint covariance; interval overlap is not the comparison test. Neither documented category establishes chronic compensation or the mechanism of acidemia.

3.5 Documented respiratory-failure acuity and OHS

This supplement partitions the inherited five-code ICD composite on the same fixed primary indexes. It requires complete encounter diagnosis capture and nonmissing inherited adjustment variables, including retained Unknown categories; it does not require blood gas evidence. J96.22 or J96.02 together with J96.12/E66.2 receives priority as **acute-on-chronic or concurrent acute/chronic coding**. Remaining J96.02 is acute; remaining J96.12/E66.2 is chronic coding/OHS; remaining J96.92 is unspecified acuity. No target code with complete capture is the reference category. Incomplete capture is unknown. OHS is grouped analytically with chronic coding and does not itself establish chronic respiratory failure.

One unpenalized five-category multinomial spline reuses the primary numeric BMI basis, linear age and inherited sex, race/ethnicity and region adjustment. All class probabilities are standardized over the same diagnosis-evaluable patients. Full joint covariance supplies pointwise 95% intervals, supported five-unit changes and pair-

wise differences in those changes. These descriptive comparisons are not multiplicity adjusted. The fitted positive components sum to the joint composite; it need not equal the existing binary model's predictions. Primary estimates, S1 and PIFs remain unchanged; no supplemental PIF is calculated.

Table 40: ICD code-level source qualification

module	status	reason	n primary complete	n complete capture	analysis population id
icd acuity v1	qualified	Complete encounter diagnosis capture; inherited five-code composite	176922	176922	primary

Table 41: ICD model estimability; source qualification does not imply a successful fit

model id	n model	status	failure reason
primary::icd acuity/ encounter/multinomial spline	176922	separation	Complete or quasi-complete multinomial separation

Adjusted ICD-acuity curves and contrasts are unavailable. The planned model was retained with its explicit failure status; no alternative estimator or recoding was substituted. The category table below is descriptive, not an adjusted BMI relationship.

Table 42: Mutually exclusive ICD categories; observed proportions with Wilson 95% intervals

Category	Patients	N	Observed	Status
No target diagnosis documented	168288	176922	95.12% (95% CI 95.02–95.22%)	qualified
Chronic coding / OHS	1769	176922	1.00% (95% CI 0.95–1.05%)	qualified
Acute-on-chronic or concurrent coding	3443	176922	1.95% (95% CI 1.88–2.01%)	qualified
Acute coding	3294	176922	1.86% (95% CI 1.80–1.93%)	qualified
Unspecified acuity	128	176922	0.07% (95% CI 0.06–0.09%)	qualified

BMI and documented respiratory-failure acuity / OHS
Primary population | Encounter-wide; shared model/reference N = 176,922
Events: chronic/OHS 1,769; acute/chronic 3,443; acute 3,294; unspecified 128

ICD components unavailable

separation: Complete or quasi-complete multinomial separation

Component intervals unavailable

separation: Complete or quasi-complete multinomial separation

No modeled comparison

separation: Complete or quasi-complete multinomial separation

Figure 11: Adjusted ICD-acuity curves and intervals are unavailable: separation: Complete or quasi-complete multinomial separation This is a model-status display, not an estimated relationship. The accompanying table reports observed category counts only when source evidence is qualified. Primary fixed indexes, adjustment and existing analyses are unchanged; no fallback estimator was used.

Table 43: ICD component magnitudes and differences in BMI-associated changes; 95% intervals

BMI interval	Outcome	Lower BMI probability % (95% CI)	Higher BMI probability % (95% CI)	Change pp (95% CI)	Probability ratio (95% CI)	Status
NA to NA	Acute-on-chronic or concurrent coding minus Chronic coding / OHS : difference in changes	Not estimated	Not estimated	Not estimated	Not estimated	separation

Table 44: Original composite and subtype reconciliation

metric	n	status	analysis population id
complete capture	176922	qualified	primary
incomplete capture	0	qualified	primary
unavailable capture	0	qualified	primary
original composite positive	8634	qualified	primary
classified positive	8634	qualified	primary

Diagnostic codes reflect documentation and historical source-entry selection. Acute coding is not acidemia, and chronic coding is not evidence of physiological compensation. OHS includes obesity in its definition; a BMI association in that component may partly reflect diagnostic incorporation. The audit retains the five-code patterns, explicit versus concurrent coding, and OHS versus chronic respiratory-failure composition by BMI and source-entry subgroup. This supplement does not establish incident disease or causal effects.

4 Discussion and example conclusion

The primary analysis estimates BMI relationships among non-OBESITY-supported indexes while preserving the same clinical episode. Excluding an obesity-only entry route does not remove outcome-dependent entry or selective gas measurement from the retained population. No weighting or transport to excluded indexes is attempted: near-absent gas observation in the excluded entry route does not supply an independent physiological comparator. The overlapping supporting analyses remain in the audit and do not constitute between-group tests.

A defensible conclusion should describe the direction and shape actually supported by the primary curves and the outcome-dependent selection exposed by the evidence audit. No claim of preventable disease or treatment benefit follows from the secondary scenario. Selection into the original files, selective gas measurement, BMI timing and measurement uncertainty, residual confounding and sparse tails remain limitations.

The accompanying audit retains full model output, reconciliation and supporting analyses. Source acceptance, numerical review, visual inspection and investigator scientific approval are distinct states.

Example wording for the revised primary population:

> **Example conclusion, pending investigator review:** We estimated BMI–hypercapnia probability relationships within the fixed-index, non-OBESITY-supported source-entry population. Outcome-specific contrasts and their uncertainty are reported separately; they do not constitute a between-group test. Gas-based estimates describe patients with usable blood-gas evidence in this selected acute-care cohort; diagnosis-based estimates describe patients with known diagnosis capture. The primary association remains subject to outcome-dependent entry and selective gas measurement. Secondary one-unit BMI-shift contrasts are descriptive model scenarios and do not establish effects of weight loss or cases prevented.